

Geospatial Foundation Models: Navigating the Trade-offs Between Scale, Efficiency, and Operational Robustness

The integration of Foundation Models (FMs) into Earth Observation signifies a paradigm shift from task-specific supervised learning to general-purpose architectures capable of leveraging vast unlabeled datasets [24]. However, this transition reveals critical tensions between scaling laws and operational reality. While billion-parameter models promise emergent abilities [OREole-FM], recent findings indicate that remote sensing FMs enter an overparameterized regime earlier than natural image counterparts, retaining high accuracy even when slimmed to 1% of their computational budget [143]. Furthermore, the necessity of domain-specific pretraining is contested, with studies showing generalist models can rival specialized backbones in certain retrieval tasks [63]. A primary challenge remains distributional robustness; benchmarks like EarthShift demonstrate that state-of-the-art FMs consistently underperform by 15-20% in out-of-distribution scenarios involving new sensors or geographies [15]. Consequently, the field is pivoting toward neuroplastic, any-sensor architectures [13] and parameter-efficient adaptation strategies like Low-Rank Adaptation to bridge the gap between benchmark performance and deployment. Future trajectories emphasize moving beyond static perception toward dynamic "world models" capable of spatiotemporal forecasting and causal reasoning [275], ensuring AI systems can robustly interpret the complex, multi-scale dynamics of our planet.

Introduction: The Paradigm Shift to Geospatial Foundation Models

The integration of Foundation Models (FMs) into Earth Observation (EO) signifies a transformative departure from the traditional reliance on task-specific, supervised algorithms toward general-purpose architectures capable of leveraging vast unlabeled datasets. This paradigm shift addresses the critical bottleneck of label scarcity, a persistent challenge in remote sensing where annotation is labor-intensive, costly, and requires specialized domain expertise [24]. By decoupling the learning of general representations from downstream task execution, FMs enable robust transferability across diverse applications, ranging from land cover mapping to disaster response, often achieving state-of-the-art performance with significantly fewer labeled samples [62]. The motivation for this transition is further underscored by the sheer volume of available data; satellites generate petabytes of imagery daily, much of which remains unlabeled, presenting an unprecedented opportunity for self-supervised learning (SSL) strategies to extract meaningful encodings without human intervention [119]. Consequently, the field is moving away from training isolated models for single tasks toward developing unified backbones that can be adapted to multiple objectives with minimal fine-tuning, thereby reducing the computational and data costs associated with traditional supervised approaches [1].

A primary driver for adopting FMs in EO is their capacity to handle the multimodal diversity inherent in geospatial data. Unlike natural image processing, which predominantly focuses on Red-Green-Blue (RGB) spectra, remote sensing encompasses a wide array of sensor modalities including optical, Synthetic Aperture Radar (SAR), LiDAR, and hyperspectral imagery, each capturing distinct physical properties of the Earth's surface [18]. Traditional models often struggle to integrate these heterogeneous data sources, yet modern FMs are increasingly designed to fuse multi-sensor inputs, leveraging complementary information to enhance representation quality. For instance, optical sensors provide rich spectral details but are hindered by cloud cover, whereas SAR offers all-weather capabilities, and their joint utilization within a unified framework improves robustness in monitoring dynamic environmental processes [64]. Furthermore, the development of models capable of processing arbitrary sensor configurations and resolutions marks a significant advancement toward "any-sensor" intelligence, allowing for seamless adaptation to new satellite missions without retraining from scratch [11]. Innovations such as dynamic hypernetworks and spectrum-independent tokenizers now enable single backbones to accommodate varying spectral bands and ground sampling distances, effectively addressing the fragmentation caused by fixed-input architectures [13] [125].

Despite these advantages, the application of FMs in EO faces unique challenges stemming from the spectral complexity and temporal dynamics of satellite data. The electromagnetic spectrum captured by remote sensing instruments extends far beyond the visible range, incorporating near-infrared, short-wave infrared, and thermal bands that require specialized architectural adaptations to process effectively [124]. Standard computer vision backbones, typically optimized for three-channel RGB inputs, often fail to fully exploit this spectral richness, necessitating the development of domain-specific tokenization and embedding strategies that preserve spectral identity [125]. Additionally, the temporal dimension introduces another layer of complexity; Earth systems are dynamic, characterized by phenological cycles, seasonal variations, and abrupt changes due to extreme events. Capturing these temporal dependencies requires models that can process irregular and unaligned time series, a capability that many existing FMs still lack, leading to potential losses in critical temporal signals when data is compositely averaged [156]. While some approaches utilize dense temporal encodings to capture short-term variability for real-time monitoring, others struggle to maintain alignment across multi-year sequences without significant architectural modifications [7].

There is also an ongoing debate regarding the necessity of domain-specific pretraining versus the utility of generalist vision models. While some studies suggest that models pretrained on natural images can compete with or even outperform EO-specific models in certain retrieval tasks, others argue that the distinct statistical properties of satellite imagery—such as the logarithmic distribution of objects and specific atmospheric effects—demand specialized pretraining objectives to achieve optimal generalization [63]. Moreover, the computational cost of training massive FMs raises concerns about sustainability and accessibility, prompting research into parameter-efficient fine-tuning methods and model slimming techniques that maintain performance while reducing resource requirements [143]. Recent findings indicate that remote sensing foundation models may enter an overparameterized regime at smaller scales than their natural image counterparts, suggesting that significant redundancy exists which can be exploited for efficient deployment [143]. The field is thus moving toward a balance between scaling laws and efficiency, exploring how to maximize the utility of FMs for open-world visual understanding while addressing the physical and operational

constraints of Earth science applications [262]. Ultimately, the evolution of geospatial FMs represents not just an incremental improvement in accuracy, but a fundamental rethinking of how AI systems perceive, reason about, and interact with the complex, multi-scale dynamics of our planet.

The Evolution from Supervised to Self-Supervised Learning

The traditional paradigm of supervised learning in remote sensing has long been constrained by the prohibitive costs and time requirements associated with generating large-scale, high-quality annotated datasets. While supervised methods have driven significant advancements in Earth Observation (EO) missions ranging from land cover classification to environmental monitoring, their reliance on human-labeled data creates a bottleneck that limits scalability and adaptability to dynamic environments [145]. This limitation is particularly acute in specialized domains such as defense-related vehicle detection or Arctic feature mapping, where expert annotation is scarce and observables are often rare [159] [22]. Consequently, the field is undergoing a profound shift toward self-supervised learning (SSL), a methodology that leverages the intrinsic structure of vast amounts of unlabeled data to learn robust feature representations without explicit human supervision [8]. By decoupling representation learning from task-specific labels, SSL enables the development of foundation models that can be effectively transferred to downstream tasks with minimal labeled examples, thereby addressing the data scarcity issue inherent in geospatial analysis [136].

Central to this evolution are sophisticated pretext tasks designed to exploit the unique characteristics of satellite imagery, specifically its rich spectral and spatial dimensions. Unlike standard computer vision applications that operate primarily in the RGB color space, remote sensing data often encompasses multispectral and hyperspectral bands that contain critical information beyond human vision. Generic SSL frameworks have been developed to specifically target these spectra-spatial features, utilizing novel pretext tasks for both object-based and pixel-based analysis that outperform methods extended directly from natural image domains [145]. Among the most prominent SSL strategies is Masked Image Modeling (MIM), including variants like Masked Autoencoders (MAE), which train models to reconstruct missing portions of an image from visible context. Research indicates that MIM-based backbones, such as Vision Transformers (ViTs) pretrained via MAE, can significantly outperform supervised counterparts in scene classification tasks, achieving accuracy gains of up to 18% on standard benchmarks [126]. Furthermore, adaptations like SatMAE++ and SIGMAE have introduced multi-scale pretraining and spectral-index-guided masking to better handle the complex backgrounds and scale variations typical of multispectral imagery, demonstrating that domain-specific modifications to masking strategies yield superior feature representations [161] [80]. Recent innovations like NeighborMAE have further extended this by exploiting spatial dependencies between neighboring images, acknowledging the continuous nature of the Earth's surface to enhance reconstruction challenges [228].

Contrastive learning represents another pillar of the SSL revolution, though it faces unique challenges in the geospatial domain regarding the definition of positive and negative pairs. Standard contrastive approaches often mistakenly treat geographically adjacent samples with similar semantic content as negative pairs, leading to training confusion. To mitigate this, innovative frameworks like SwiMDiff have integrated scene-wide matching with diffusion constraints to recalibrate labels and capture fine-grained details, while others like Multi-label Guided Soft Contrastive Learning utilize free global semantic information to resolve bottlenecks in positive sample matching [247] [248]. Additionally, the emergence of diffusion models has opened new avenues for generative pretraining, with studies showing that generative geospatial diffusion models can rival or even exceed discriminative foundation models in tasks such as semantic segmentation and classification [68] [134].

The efficacy of SSL pretraining on domain-specific data versus general-purpose datasets like ImageNet remains a subject of rigorous investigation and debate. While a common assumption posits that domain-aligned pretraining offers maximal benefits, systematic studies reveal a nuanced reality. Some research suggests that SSL pretraining on remote sensing data provides only modest improvements over ImageNet pretraining in certain few-shot settings, questioning whether the additional costs of data curation are always justified [138] [289]. However, contrasting evidence from specialized applications demonstrates that in-domain SSL models significantly outperform ImageNet baselines, particularly in low-label regimes and for tasks requiring specific spectral understanding, such as SAR data analysis or hyperspectral classification [159] [105] [124]. For instance, models pretrained on curated Arctic imagery have shown substantial improvements over general Earth observation foundation models, highlighting the value of region-specific data distribution optimization [22]. Moreover, the integration of temporal dynamics and multi-modal data through frameworks like TESSERA and MAESTRO further underscores the potential of SSL to capture complex spatio-temporal patterns that supervised methods often miss due to compositing or single-modality limitations [212] [95]. Ultimately, the transition to self-supervised strategies facilitates the creation of versatile foundation models

capable of generalizing across diverse geospatial tasks, sensors, and geographic regions, marking a definitive paradigm shift in how Earth observation data is utilized for scientific and operational purposes [9] [53].

Multimodality, Sensor Diversity, and Data Heterogeneity

The paradigm shift toward geospatial foundation models is fundamentally driven by the necessity to transcend the limitations of single-sensor architectures, moving instead toward unified frameworks capable of ingesting and synthesizing heterogeneous data streams. Earth observation data is inherently diverse, characterized by varying spectral bands, spatial resolutions, and sensing modalities ranging from optical and Synthetic Aperture Radar (SAR) to LiDAR and hyperspectral imagery. Traditional models, often constrained to fixed input configurations, struggle to generalize across the dynamic landscape of satellite missions where band compositions and ground sampling distances differ significantly. To address this, recent research has prioritized the development of "any-sensor" models that treat sensor diversity not as a barrier, but as a feature to be encoded within a shared representation space. For instance, the Any-Optical-Model introduces a spectrum-independent tokenizer and multi-scale adaptive patch embedding to accommodate arbitrary band compositions and resolution scales, effectively handling missing bands and cross-sensor fusion without retraining [125]. Similarly, Panopticon advances this concept by treating co-located images from different sensors as natural augmentations and employing cross-attention over channels to process arbitrary combinations of optical and SAR inputs, thereby achieving sensor-agnostic generalization [11].

A critical challenge in unifying these modalities lies in encoding the distinct physical properties of each sensor type while maintaining semantic consistency. The Dynamic One-For-All (DOFA) model draws inspiration from neural plasticity, utilizing a wavelength-conditioned dynamic hypernetwork to generate weights specific to the input modality, allowing a single backbone to adapt flexibly to unseen spectral configurations [13]. This approach contrasts with methods that rely on separate encoders for each modality, which often fail to capture inter-sensor relationships efficiently. Furthermore, the integration of hyperspectral imagery (HSI), which offers rich spectral detail but suffers from limited coverage, remains a frontier for multimodal fusion. SpectralEarth-FM addresses the underrepresentation of HSI in multimodal pretraining by introducing a hierarchical transformer that jointly processes hyperspectral data with multispectral, SAR, and land surface temperature inputs, leveraging a Joint-Embedding Predictive Architecture to align representations across these heterogeneous sources [72]. Complementary efforts, such as the synthesis of hyperspectral data from multispectral inputs via spectral transformers, demonstrate how bridging the trade-off between spectral resolution and spatial coverage can enhance specific applications like greenhouse gas estimation [251].

Despite these architectural advancements, the practical deployment of multimodal models is hindered by the lack of standardized access to pre-computed embeddings and the fragmentation of data formats. The ecosystem of geospatial foundation models currently suffers from incompatible interfaces and varying resolution standards, creating an engineering bottleneck for practitioners. Initiatives like TESSERA aim to resolve this by providing ready-to-use, global annual embeddings at 10-meter resolution derived from fused optical and SAR time series, eliminating the need for users to perform computationally intensive inference or data processing [212]. However, the scarcity of standardized, analysis-ready products persists, prompting calls for a unified taxonomy and API to treat embeddings as first-class geospatial datasets [128]. Additionally, the heterogeneity of data extends to temporal alignment; models like ALISE and MAESTRO have been developed specifically to handle irregular and unaligned satellite image time series, ensuring that temporal dynamics are preserved even when sampling rates vary across sensors [156] [95].

The complexity of data heterogeneity also necessitates innovative strategies for handling resolution disparities and missing modalities during inference. Approaches such as THOR introduce "compute-adaptive" architectures capable of processing native resolutions from 10 meters to 1000 meters within a single model, enabling dynamic trade-offs between computational cost and feature resolution [27]. In scenarios where specific modalities are unavailable, training-free methods have emerged to adapt generalist large multimodal models to multispectral inputs by injecting domain-specific instructions and adapting non-RGB inputs to the visual space understood by RGB-trained models [176] [54]. Furthermore, the integration of non-raster data, such as vector semantics from OpenStreetMap, with raster imagery represents a growing direction for creating human-centric geospatial models that unify discrete object relationships with continuous physical patterns [114]. Ultimately, the trajectory of geospatial foundation models points toward increasingly flexible, sensor-agnostic systems that can leverage the full spectrum of available Earth observation data, from thermal CubeSat imagery to high-resolution commercial optical data, as evidenced by datasets like SeeFar and TerraMesh which curate multi-resolution, multi-modal samples for robust pretraining [73] [64].

Challenges in Generalization, Robustness, and Operational Reality

The transition from benchmark-driven research to operational deployment in Earth Observation (EO) reveals a critical disconnect between the reported performance of Geospatial Foundation Models (GFM) and their reliability in real-world scenarios. A primary obstacle is the prevalence of distribution shifts, where models trained on specific geographies, time periods, or sensor configurations fail to generalize to out-of-distribution (OOD) contexts. Recent benchmarking efforts have quantified this fragility, demonstrating that even state-of-the-art GFM suffer significant performance degradation when faced with realistic shifts. For instance, experiments across multiple tasks and shift types indicate that GFM consistently perform 15-20% worse in OOD scenarios compared to in-distribution settings, a deficit that persists regardless of model architecture, size, or pre-training strategy [15]. This lack of robustness is not unique to geospatial models; their resilience mirrors that of generic vision foundation models, suggesting a systemic issue in how current architectures learn invariant features from heterogeneous EO data. Furthermore, the challenge of open-world recognition remains acute, as deployed models must continuously adjust to a constant influx of new data exhibiting diverse semantic and covariate shifts, a capability that existing single-dataset evaluations often fail to capture [56].

The challenge of generalization is further compounded by the diversity of sensor modalities and the dynamic nature of satellite constellations. As new satellites are launched with varying spectral bands and spatial resolutions, models trained on legacy data often struggle to adapt without extensive retraining. Studies show that when generalizing to satellites with no band overlap, performance can drop by a factor of two to four, highlighting the limitations of fixed-input architectures [38]. While some models attempt to address this through any-sensor designs that process arbitrary channel combinations, they still face difficulties in maintaining consistent accuracy across unseen sensor configurations [11]. To overcome this rigidity, neuroplasticity-inspired architectures like DOFA utilize wavelength-conditioned dynamic hypernetworks to flexibly process inputs from distinct satellite sensors, enabling generalization to unseen modalities without retraining [13]. Similarly, compute-adaptive models like THOR allow a single set of weights to operate at varying patch sizes and resolutions, addressing the tension between input heterogeneity and deployment rigidity [27]. The geographic distribution of pre-training data also plays a pivotal role; models trained on region-specific data often fail to capture global patterns, whereas balanced, globally representative datasets tend to yield better generalization capabilities [144]. Consequently, the bottleneck often lies not solely in the model architecture but in the data distribution shifts and the lack of representative training samples, necessitating a shift toward data-centric approaches that identify necessary and sufficient features for optimal performance [234].

Beyond algorithmic robustness, operational reality imposes stringent constraints on latency, computational resources, and energy efficiency that are rarely reflected in academic benchmarks. Operational systems frequently run on resource-constrained hardware, such as onboard satellite processors or edge devices, where the massive computational footprint of large GFM becomes a prohibitive barrier. To bridge this gap, researchers are exploring model compression techniques, including knowledge distillation and parameter-efficient fine-tuning methods like Low-Rank Adaptation (LoRA). These approaches have shown promise in reducing model size and computational costs while maintaining competitive performance, enabling the deployment of capable models on flight-ready hardware [57] [116]. For example, compact variants of Vision Transformers have been successfully demonstrated on the International Space Station, proving that high-performance inference is feasible under strict power and memory constraints [57]. Similarly, dynamic weighting knowledge distillation frameworks allow lightweight student models to achieve high accuracy with significantly fewer parameters and FLOPs, making them suitable for onboard classification tasks [88]. Moreover, recent analyses suggest that remote sensing foundation models operate in an overparameterized regime where significant redundancy exists, allowing models to be slimmed post-hoc with minimal performance loss, thus offering a practical pathway for resource-constrained deployment [143].

However, simply compressing models does not fully resolve the tension between flexibility and efficiency. Operational environments often require dynamic trade-offs between compute cost and feature resolution, necessitating architectures that can adapt their inference strategy in real-time. The integration of edge intelligence within satellite constellations offers a pathway to reduce downlink latency and bandwidth usage by performing semantic processing onboard. By optimizing observation and processing schedules jointly, these systems can prioritize high-value data transmission and enable near real-time monitoring for time-critical applications like disaster response [173] [160]. Yet, the deployment of

such systems requires a structured protocol that moves beyond accuracy metrics to include requirements for reliability, ease of integration, and adherence to legacy data pipelines [104]. The disconnect between research and operations is also evident in agricultural and ecological applications, where foundation model embeddings often exhibit limited spatial transferability and struggle with cross-country generalization when faced with distinct yield distributions or phenological cycles [89] [146]. Consequently, future research must prioritize the development of robust, resource-efficient, and adaptable models that can withstand the rigors of operational deployment, moving away from a sole focus on benchmark scores toward holistic system design that accounts for the full spectrum of real-world challenges [43].

Architectural Innovations and Scaling Strategies

The architectural landscape for Earth Observation (EO) has evolved from the direct adaptation of generic Vision Transformers (ViTs) to the design of specialized backbones that explicitly address the unique physical and operational constraints of geospatial data. While standard ViTs offer strong scalability, their application in remote sensing requires modifications to handle arbitrary object orientations, vast spatial scales, and multi-spectral inputs without incurring prohibitive computational costs [254]. To mitigate the quadratic complexity of full attention mechanisms, researchers have introduced rotated varied-size window attention, which significantly reduces memory footprint while enhancing object representation by extracting rich context from diverse windows tailored to oriented objects [254]. Furthermore, lightweight variants like VistaFormer have demonstrated that substituting Multi-Head Self-Attention with Neighbourhood Attention can achieve superior segmentation performance for satellite image time series while utilizing only a fraction of the floating-point operations required by standard transformers [185]. These adaptations highlight a shift toward efficiency-aware designs that maintain high accuracy for tasks such as crop type segmentation while drastically reducing resource demands [185].

A critical frontier in this evolution is the integration of State Space Models (SSMs), such as Mamba, to overcome the inherent scaling limitations of attention-based architectures. The PathMamba architecture exemplifies this hybrid approach, strategically combining Mamba blocks for tracing continuous road networks with Transformer blocks for global reasoning, thereby achieving topological coherence without the prohibitive scaling costs of pure attention models [233]. This shift toward linear-complexity architectures is further validated by RoMA, a framework enabling scalable self-supervised pretraining of Mamba-based foundation models on large-scale unlabeled data [23]. RoMA demonstrates that Mamba architectures adhere to remote sensing scaling laws, consistently outperforming ViT-based counterparts in both accuracy and computational efficiency across detection and segmentation tasks [23]. Similarly, SatMamba proposes a pretraining framework combining masked autoencoders with State Space Models to address the limitations of ViTs in processing long sequences of multiband and multitemporal data, offering a pathway to more efficient foundation models [16].

Beyond backbone modifications, scaling strategies increasingly focus on handling multi-modal and multi-scale data through dynamic and unified architectures. The Dynamic One-For-All (DOFA) model introduces a neural plasticity-inspired design utilizing wavelength-conditioned dynamic hypernetworks, allowing a single unified Transformer backbone to flexibly adapt to diverse sensor modalities and unseen spectral bands [13]. Complementing this, THOR presents a "compute-adaptive" foundation model capable of processing native resolutions from multiple Copernicus satellites within a single architecture, enabling dynamic trade-offs between computational cost and feature resolution without retraining [27]. The push for unification is also evident in OFA-Net, which employs a single shared Transformer backbone for multiple data modalities with different spatial resolutions, eliminating the need to switch backbones based on input data [87]. These approaches address the rigidity of models tailored to specific sensors, fostering a more adaptable ecosystem for open-world EO applications [13].

The relationship between model size and performance remains a subject of rigorous investigation, with evidence supporting the benefits of billion-scale parameters yet challenging the necessity of indiscriminate growth. Studies indicate that increasing parameter counts from millions to billions yields improved performance and data efficiency in downstream tasks such as rotated object detection and semantic segmentation [112]. Efforts like OReole-FM leverage exascale computing to pretrain billion-parameter models, highlighting the importance of data scaling alongside model scaling to unlock emergent abilities [10]. However, this trend faces scrutiny regarding redundancy; research suggests that remote sensing foundation models may enter an overparameterized regime at smaller scales than their computer vision counterparts, distributing task-relevant information with high redundancy [143]. Consequently, post-hoc slimming techniques can reduce computational budgets significantly while retaining over 71% relative accuracy, challenging the prevailing scaling paradigm and offering a practical strategy for resource-constrained deployment [143].

Hybrid architectures combining Convolutional Neural Networks (CNNs) and Transformers continue to offer viable solutions for specific constraints, particularly in onboard processing where energy efficiency is paramount. Comparative studies for land use classification indicate that efficient ViT variants like EfficientViT-M2 often outperform traditional CNNs and larger Transformers in terms of robustness to noise and energy efficiency, making

them optimal for satellite operations [55]. Nevertheless, ensemble approaches that combine slow-to-train, high-accuracy ViTs with fast-to-train CNNs provide a scalable strategy for handling incremental data volumes, balancing accuracy with the cost of retraining in big data contexts [157]. Additionally, hybrid CNN-Transformer structures have proven effective in power plant detection, integrating the local feature extraction of CNNs with the long-range dependency capture of ViTs to enhance monitoring capabilities [162]. These diverse architectural trajectories underscore a field moving toward specialized, efficient, and adaptable models rather than a one-size-fits-all scaling paradigm, ensuring that future systems can operate effectively across the full spectrum of EO challenges.

Transformer Variants, State Space Models, and Efficient Backbones

The dominance of Transformer-based backbones in remote sensing has been tempered by the computational bottleneck of self-attention, which scales quadratically with sequence length, rendering it prohibitive for ultra-high-resolution imagery and long temporal sequences. To overcome this, the field has pivoted toward hierarchical adaptations and linear-complexity alternatives that preserve global context while drastically reducing memory footprints. Hierarchical Vision Transformers have become pivotal in processing high-resolution satellite data by limiting attention computation to local neighborhoods while maintaining global connectivity through deeper layers. This approach is exemplified by architectures that extend Masked Autoencoders to integrate temporal information, employing hierarchical 3D blocks to effectively capture multi-scale spatio-temporal dependencies for tasks like land cover segmentation and flood mapping [21]. Similarly, lightweight adaptations have demonstrated that substituting standard Multi-Head Self-Attention with Neighbourhood Attention can achieve comparable or superior accuracy in satellite image time series segmentation while requiring only a fraction of the floating-point operations, highlighting the efficacy of localized attention mechanisms in reducing computational overhead without sacrificing performance [185]. Further refining this local-global balance, recent works have introduced rotated varied-size window attention to handle objects of arbitrary orientations, significantly reducing computational costs while learning richer object representations for detection tasks [254].

Parallel to these Transformer adaptations, State Space Models (SSMs), particularly the Mamba architecture, have emerged as a promising alternative offering linear computational scaling with sequence length. This characteristic makes them uniquely suited for processing the extremely long sequences generated by high-resolution remote sensing data. Recent frameworks have successfully scaled up Mamba-based foundation models through self-supervised pretraining on large-scale unlabeled datasets, validating that Mamba adheres to scaling laws in the remote sensing domain and consistently outperforms ViT-based counterparts in both accuracy and efficiency across detection and segmentation tasks [23]. Complementing this, other proposals combine masked autoencoders with SSMs to handle multiband and multitemporal data, aiming to unlock the full potential of Earth observation data by bypassing the quadratic scaling limits of traditional attention [16]. However, recognizing that pure SSMs and pure Transformers possess complementary strengths, several studies advocate for hybrid architectures. For instance, novel designs integrate Mamba's sequential modeling capabilities with the global reasoning of Transformers specifically for road segmentation, where Mamba blocks trace the continuous, topological nature of road networks while Transformer blocks refine features with global context, yielding topologically superior segmentation maps without prohibitive costs [233].

Beyond the core backbone mechanics, efficiency is also pursued through novel attention biases and tokenization strategies that enhance scalability across varying resolutions and modalities. Innovations such as extending linear bias attention mechanisms to incorporate Ground Sample Distance (GSD) scaling allow transformers to natively process multi-scale and multi-modal imagery by encoding spatial relationships between patches of different resolutions, facilitating the comparison of images at different scales within a single model [82]. Furthermore, low-rank efficient spatial-spectral attention blocks have been introduced to approximate high-dimensional attention, significantly improving computational efficiency for hyperspectral and multi-modal geospatial data while maintaining competitive performance on cross-satellite generalization tasks [182]. These efficient backbones are increasingly critical as benchmarks begin to evaluate model capabilities on extremely large, ultra-high-resolution imagery, where standard tokenization strategies fail to capture long-range dependencies without excessive computational burden [194]. The convergence of hierarchical designs, linear-complexity SSMs, and efficient attention mechanisms marks a critical evolution in creating backbones capable of handling the increasing volume and resolution of modern satellite data, moving the field away from brute-force scaling toward mathematically grounded modifications that ensure feasibility for operational Earth observation tasks.

Dynamic, Neuroplastic, and Hybrid Architectures

The evolution of Earth observation (EO) foundation models has necessitated a paradigm shift from rigid, sensor-specific architectures toward dynamic systems capable of mimicking biological neuroplasticity to handle the open-world nature of satellite data. Traditional models often struggle with the heterogeneity of EO data, where varying spectral bands, spatial resolutions, and sensor modalities typically require distinct processing pipelines or separate encoders, leading to significant computational inefficiencies and limited generalization. To address this inflexibility, recent innovations have introduced architectures that dynamically adjust their internal weights based on input characteristics, effectively creating a "one-for-all" framework. A prime example of this neuroplasticity-inspired approach is the Dynamic One-For-All (DOFA) model, which utilizes a wavelength-conditioned dynamic hypernetwork to generate network weights tailored to the specific central wavelengths of input spectral bands [13]. This mechanism allows the model to flexibly accommodate unseen sensors and varying spectral configurations without retraining, emulating the brain's capacity to reorganize in response to novel stimuli. Similarly, the Any Optical Model (AOM) advances this concept by employing a spectrum-independent tokenizer and a multi-scale adaptive patch embedding mechanism, enabling the processing of arbitrary band compositions and resolution scales while preserving distinctive spectral identities [125]. These approaches contrast sharply with earlier methods that relied on fixed spectral encodings, thereby overcoming the limitations that previously hindered generalization across the diverse EO landscape [13]. Furthermore, the SenPa-MAE architecture explicitly encodes sensor parameters, such as ground sampling distance and spectral response functions, into the image embeddings, allowing a single model to differentiate between and effectively process data from diverse satellite missions with non-matching characteristics [191].

Parallel to dynamic weight generation, the integration of hybrid architectures has emerged as a critical strategy to leverage the complementary strengths of Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), addressing the specific needs of different EO tasks. While CNNs excel at extracting local features and textures, ViTs are superior in modeling long-range dependencies and global context, yet neither is universally optimal for all scenarios. Research indicates that hybrid models combining these paradigms often outperform their singular counterparts in complex detection tasks. For instance, a hybrid architecture integrating CNN feature extraction with ViT global reasoning demonstrated improved accuracy in power plant detection and energy estimation compared to standalone models [162]. However, the superiority of hybrid or transformer-based approaches is not absolute; in wildfire detection scenarios, while ViTs showed competitive performance against specialized CNNs, well-tuned CNN-based U-Nets still achieved the highest Intersection over Union (IoU) scores, suggesting that a synergistic combination of local and global processing remains optimal for certain high-precision tasks [243]. This duality is also evident in road segmentation, where the PathMamba architecture hybridizes State Space Models (Mamba) with Transformers to ensure both topological continuity and global contextual understanding, addressing the quadratic complexity limitations of pure attention mechanisms [233]. Additionally, foundational work on the Netherlands utilized a combined CNN-ViT structure to capture both low-frequency landscape features and high-frequency textures, proving effective even with limited pretraining data [44]. The emergence of State Space Models like Mamba offers a promising alternative to standard attention, as seen in SatMamba, which combines masked autoencoders with State Space Models to offer linear computational scaling for multiband and multitemporal data [16].

Beyond architectural hybridity, compute-adaptive designs have been developed to handle the ingestion of native resolutions from multiple sensors simultaneously, addressing the computational inefficiencies of resizing or resampling data which often degrades model performance. The THOR model represents a significant advancement in this domain as a "compute-adaptive" foundation model capable of unifying data from Copernicus Sentinel-1, -2, and -3 satellites, processing their native resolutions ranging from 10 meters to 1000 meters within a single framework [27]. By employing a randomized patch and input image size strategy during pretraining, THOR enables dynamic trade-offs between computational cost and feature resolution at inference time without requiring retraining, a flexibility crucial for real-world deployment where resource constraints vary. Complementing this, the Panopticon model treats co-located images from different sensors as natural augmentations and incorporates cross-attention over channels to process arbitrary sensor combinations, effectively functioning as an any-sensor foundation model [11]. These compute-adaptive and sensor-aware strategies collectively reduce the need for multiple specialized models, fostering a more sustainable and scalable ecosystem for Earth observation analytics. The convergence of these dynamic, hybrid, and adaptive architectures signifies a move toward open-world EO visual understanding, where frameworks can interpolate in weight

space, fuse multi-scale features, and adapt to computational constraints on the fly, unlocking new capabilities in monitoring global challenges by providing robust, generalizable, and efficient solutions for heterogeneous data environments.

Generative, Decoder-Only, and Diffusion-Based Approaches

The architectural landscape of remote sensing foundation models is undergoing a paradigmatic shift, moving away from strictly encoder-only discriminative frameworks toward generative, decoder-only, and diffusion-based architectures inspired by the scalability of Large Language Models. This evolution addresses critical limitations inherent in deterministic models, particularly the "regression to the mean" problem where minimizing pixel-based errors results in blurry outputs that obscure subtle geographic-spatial features. To mitigate this, recent works propose hybrid frameworks that combine Self-Supervised Learning (SSL) with Hidden Diffusion Models to balance structural accuracy with textural fidelity. For instance, by predicting stable semantic representations via an IJEPa module and routing them through a frozen Stable Diffusion backbone, models can synthesize realistic textures while ensuring absolutely accurate structural predictions, significantly outperforming deterministic baselines in perceptual metrics [134].

Diffusion models have emerged as a cornerstone for generative tasks, offering capabilities that extend beyond simple image synthesis to serve as powerful discriminative backbones. Unlike standard diffusion models conditioned on text captions, which are sparse for satellite imagery, specialized approaches leverage metadata such as geolocation to support complex conditional generation tasks including temporal generation, super-resolution, and in-painting [187]. Furthermore, the utility of diffusion models is not limited to generation; recent frameworks demonstrate that generative diffusion models can be transformed into potent discriminative tools by systematically analyzing multi-stage, noise-dependent diffusion features. This approach challenges the notion that discriminative objectives are superior for representation learning, achieving state-of-the-art performance in semantic segmentation and classification by leveraging the multigrained semantics captured during the generative process [68].

Parallel to the rise of diffusion models, there is a growing interest in decoder-only architectures that treat Earth Observation data as unified token sequences, mirroring the next-token prediction strategy of LLMs. SITS-DECO introduces a proof-of-concept model that applies a GPT-style decoder-only architecture to Satellite Image Time Series, framing diverse tasks ranging from pixel-wise classification to multi-temporal analysis within a single generative framework without task-specific adaptations [45]. This approach highlights that dense temporal sequence modeling is a critical component often missing in current encoder-centric paradigms. Similarly, EarthPT presents a 700 million parameter decoding transformer trained in an autoregressive manner to forecast future pixel-level surface reflectances, suggesting that the abundance of EO data allows for scaling laws similar to those observed in LLMs with no immediate data-imposed limits to model growth [117].

The application of these generative frameworks extends to highly specific conditional tasks, such as map-to-image translation and global-scale simulation. Pretrained diffusion models can be conditioned on cartographic data to generate realistic satellite views, maintaining high fidelity to the underlying map structure, as demonstrated in studies focusing on region-specific generation [26]. On a broader scale, MetaEarth breaks barriers in image generation scale by proposing a resolution-guided self-cascading framework, enabling the creation of unbounded, arbitrary-sized images at various geographical resolutions to serve as a data engine for downstream tasks [79]. Additionally, generative models are proving essential for data imputation and modality translation, such as synthesizing optical images from SAR inputs to address data gaps caused by cloud cover, where generated images provide valuable information for downstream classification despite variations in perceptual quality [273].

Despite the promise of these architectures, challenges remain regarding computational efficiency and the integration of multi-modal data. While Vision Transformers have been dominant, their quadratic complexity poses scalability issues for high-resolution imagery. Consequently, emerging State Space Models like Mamba are being integrated into hybrid architectures to achieve linear-time efficiency while preserving global context, as seen in models designed for topologically coherent road segmentation and scalable foundation pretraining [233] [23]. Furthermore, the debate continues regarding the necessity of domain-specific pretraining versus general-purpose models, with some studies challenging the assumption that specialized foundation models always outperform general-purpose baselines at smaller scales [29]. Nevertheless, the consensus indicates that generative and decoder-only approaches offer a more flexible and robust foundation for the future of geospatial AI, capable of handling the spatio-temporal complexity inherent in Earth Observation data.

Scaling Laws, Compute Efficiency, and Model Redundancy

The trajectory of foundation model development in remote sensing is currently undergoing a critical re-evaluation of scaling laws, revealing a distinct divergence from the paradigms established in general computer vision and natural language processing. While domains like natural language often exhibit predictable performance gains that scale with parameter count, empirical evidence in remote sensing suggests that models frequently enter an overparameterized regime at substantially smaller scales. Research utilizing petascale electro-optical datasets has observed that increasing Vision Transformer backbones from 86 million to 1.8 billion parameters can yield effectively flat returns in performance, indicating that model scaling alone provides negligible benefits under typical data size regimes [12]. This phenomenon contrasts with the expectation that larger models inherently unlock richer context-aware representations, a discrepancy attributed to the finite variety of observable objects and the intrinsic spatial resolution limits of overhead imagery [12]. However, this saturation is not absolute; other investigations utilizing exascale computing resources argue that billion-scale models can still yield emergent abilities and significant performance boosts, provided that data scaling and high-resolution inputs are rigorously matched to the model capacity [10]. Consequently, the prevailing assumption that larger models invariably yield more universal representations requires significant qualification in the geospatial domain, where the interplay between data diversity and model capacity is more complex than simple parameter accumulation.

A central concept emerging from this re-evaluation is "slimmability," which posits that remote sensing foundation models possess high representational redundancy. Investigations into post-hoc slimming, involving the uniform reduction of pretrained encoder widths, have demonstrated that remote sensing models can retain over 71% relative accuracy at just 1% of their original computational budget [143]. This stands in stark contrast to computer vision counterparts, such as Masked Autoencoders trained on ImageNet, which suffer catastrophic performance degradation under similar compression, retaining less than 10% accuracy [143]. This sevenfold difference in robustness implies that remote sensing models distribute task-relevant information across many interchangeable parameters, creating an "implicit ensemble" rather than encoding features sparsely [143]. Mechanistic analyses using explained variance ratios and feature correlation confirm that this redundancy allows for significant parameter reduction without the need for structured, task-aware pruning, challenging the necessity of massive parameter counts for effective representation learning in this domain [143].

In response to these diminishing returns on parameter scaling, the community is shifting focus toward optimizing compute budgets through architectural innovations and lightweight network design. Strategies such as Low-Rank Adaptation (LoRA) have proven highly effective for adapting large foundation models to specific tasks while updating less than 1% of parameters, offering a favorable trade-off between accuracy and efficiency in scenarios involving geographic and temporal domain shifts [240] [116]. Furthermore, the development of slimmable networks that can operate at multiple widths facilitates dynamic accuracy-efficiency trade-offs at runtime, a capability particularly vital for resource-constrained environments such as onboard satellite systems [143]. Frameworks like InstaGeo exemplify this trend by employing task-specific distillation to transform large foundation models into student models that are up to eight times smaller, significantly reducing inference FLOPs and carbon emissions with minimal performance degradation [43]. Similarly, lightweight architectures such as MKANet and VistaFormer utilize shallow designs and neighborhood attention mechanisms to achieve state-of-the-art segmentation results with a fraction of the computational cost of standard transformers [109] [185].

The interplay between data diversity and model capacity further complicates the scaling landscape. While some studies argue that data scaling remains the primary driver of performance, others suggest that the spatial distribution and diversity of pre-training data are more critical than sheer volume or model size [144] [12]. The construction of diverse, multi-modal datasets aims to address data limitations, yet findings indicate that even with massive datasets, the benefits of scaling model parameters eventually saturate without careful curation [73] [4]. This saturation point suggests that future advancements may rely less on brute-force parameter increases and more on efficient data curation, dynamic dataset pruning to maximize diversity, and the development of compute-adaptive architectures like THOR that can handle heterogeneous sensor inputs without retraining [69] [27]. Ultimately, the path forward involves a strategic balance where model complexity is aligned with task difficulty and data availability, moving away from a singular focus on scale toward sustainable, efficient, and adaptable geospatial intelligence.

Pretraining Data Ecosystems, Curation, and Standardization

The efficacy of Geospatial Foundation Models (GFM) is increasingly recognized as being contingent not merely on architectural sophistication, but fundamentally on the scale, diversity, and strategic curation of their pretraining data ecosystems. While early research prioritized novel pretext tasks and transformer variants, recent empirical evidence underscores that the geographic and spectral distribution of pretraining data is a critical determinant of model generalization. Studies indicate that balanced, globally representative data compositions often outperform region-specific sampling, although the optimal strategy remains dependent on the specific GFM architecture employed [144]. This data-centric shift has catalyzed the development of massive, multimodal datasets designed to capture the heterogeneity of Earth observation signals. For instance, TerraMesh introduces a globally diverse dataset combining optical, synthetic aperture radar (SAR), elevation, and land-cover modalities in an Analysis-Ready Data format, significantly expanding the sample count available for large-scale pretraining compared to predecessors like So2Sat or BigEarthNet-MM [64]. Similarly, updates to datasets like SSL4EO-S12 v1.1 address previous limitations in data alignment by packaging nearly one million patches covering the world's largest cities across four seasons in a cloud-efficient Zarr format, thereby facilitating low-barrier access for self-supervised learning [229].

Beyond sheer volume, the strategic curation of data to address specific domain biases has emerged as a paramount concern. Standard global datasets frequently underrepresent ecological regions or fail to capture local phenological cycles, leading to models biased toward areas of high human activity. To mitigate this, SSL4Eco proposes a phenology-informed sampling strategy that consistently improves representation quality for ecological downstream tasks compared to traditional calendar-based sampling [91]. Furthermore, dynamic dataset pruning strategies have been introduced to enhance training efficiency by maximizing diversity and reducing redundancy in heavy-tailed distributions, as demonstrated in the Sentinel-1 Wave Mode archive [69]. In specialized domains such as the Arctic, clustering-guided curation of very high-resolution imagery has proven superior to general-purpose pretraining, yielding significant performance gains in fine-scale feature mapping [22]. These findings collectively challenge the notion that "more data is always better," suggesting instead that targeted curation and feature selection can achieve optimal performance with a fraction of the temporal instances or spectral bands [234].

A parallel evolution in the data ecosystem involves the standardization of data formats and the treatment of embeddings as first-class geospatial products. The fragmentation of embedding formats and resolutions has created engineering bottlenecks that hinder reproducibility and model comparison. Initiatives like Major TOM aim to standardize open, AI-ready datasets, while recent work formalizes a taxonomy to treat pre-computed embeddings as interoperable data products, decoupling downstream analysis from model-specific engineering [128]. Tools such as TorchGeo extensions and libraries like rs-embed provide unified APIs for loading and querying diverse embedding products, enabling users to retrieve embeddings for any location and time range with minimal friction [203]. This standardization extends to the creation of analysis-ready data cubes, such as the Agriculture monitoring Data Cube (ADC), which automates the preprocessing and indexing of optical and SAR images into multidimensional structures suitable for time-series analysis [278]. Moreover, models like TESSERA now generate ready-to-use, gap-free global embedding maps, effectively removing the computational burden of inference for end-users [212].

The integration of heterogeneous data sources remains a central challenge, driving the development of models capable of handling missing modalities and varying resolutions. Frameworks like the Earth System Foundation Model (ESFM) extend training protocols to handle diverse datasets, including sparse station data and gridded reanalysis, under a single backbone by utilizing axial attention to capture inter-variable dependencies [93]. Similarly, THOR introduces a "compute-adaptive" architecture capable of processing native resolutions from multiple Copernicus sensors simultaneously, allowing for dynamic trade-offs between computational cost and feature resolution [27]. The push toward sensor-agnostic modeling is further exemplified by Panopticon and SeeFar, which train on arbitrary sensor combinations and multi-resolution imagery to ensure generalization across existing and future satellite missions [11; 73]. Moreover, the fusion of raster imagery with vector semantics, such as OpenStreetMap data, is emerging as a critical direction for developing human-centric models that leverage explicit geographic relationships alongside continuous spectral patterns [114].

Despite these advances, significant gaps persist regarding robustness to distribution shifts and the efficiency of model adaptation. Benchmarks like EarthShift reveal that GFMs consistently suffer performance degradation when faced with

out-of-distribution scenarios involving new geographies, time periods, or sensors, highlighting the need for future research to prioritize robustness over mere accuracy [15]. Additionally, while scaling model parameters is a common strategy, studies indicate that remote sensing foundation models may enter an overparameterized regime sooner than their computer vision counterparts, suggesting that post-hoc slimming and parameter-efficient fine-tuning methods like Low-Rank Adaptation (LoRA) offer viable pathways for deployment in resource-constrained environments without substantial performance loss [143; 240]. The ecosystem is thus maturing from a focus on raw data accumulation to a sophisticated interplay of curated diversity, standardized access, and efficient, robust utilization.

Global Multi-Sensor and Multi-Temporal Datasets

The evolution of pretraining data ecosystems in Earth observation has shifted decisively from regionally constrained, single-sensor collections toward massive, globally extensive datasets that integrate multiple sensor modalities and dense temporal dimensions. This transition is driven by the recognition that generalizable representations require exposure to the full spectrum of environmental variability, including seasonal cycles, diverse geographic biomes, and distinct physical sensing mechanisms. Early efforts often relied on datasets limited to specific urban centers or single seasons, but recent initiatives have prioritized the curation of analysis-ready data that spans the globe to overcome the limitations of ImageNet-based pretraining. A pivotal development in this domain is the SSL4EO family, which established benchmarks for self-supervised learning using Sentinel imagery. The original SSL4EO-S12 introduced a large-scale, multi-modal collection designed to capture seasonal dynamics across global cities, moving beyond the constraints of labeled data scarcity [85]. Recognizing the need for improved data alignment and cloud-efficient access, subsequent iterations like SSL4EO-S12 v1.1 expanded coverage to include the world's 10,000 largest cities and their surroundings, packaging nearly one million patches in Zarr format to facilitate low-barrier processing and standardize metadata inclusion [229]. Parallel to optical-focused efforts, SSL4EO-L addressed the historical underutilization of the Landsat archive by creating the largest Landsat dataset in history, comprising 5 million image patches across multiple sensors and product levels, thereby enabling the first foundation models specifically tailored to the unique spectral characteristics of the Landsat program [96].

Beyond the SSL4EO series, other initiatives have pushed the boundaries of scale and modality integration to capture broader planetary patterns and address geographic biases inherent in earlier collections. MajorTOM represents a significant leap in spatial coverage, offering a 23TB unlabeled dataset that encompasses the vast majority of the Earth's surface, allowing for the pretraining of models like PhilEO on a truly global scale rather than being restricted to urban centers [4]. This expansion is critical, as studies indicate that models trained on data-rich developed nations may struggle to transfer to data-scarce regions, highlighting the importance of truly global sampling strategies [235]. Similarly, SatlasPretrain combines Sentinel-2 and high-resolution NAIP imagery with hundreds of millions of labels, creating a resource that bridges the gap between medium-resolution global monitoring and fine-grained local analysis, demonstrating substantial performance gains over general vision baselines [19]. The drive for multimodal fusion is further exemplified by TerraMesh, which integrates optical, synthetic aperture radar (SAR), elevation, and land-cover modalities into a single analysis-ready framework. With over 9 million samples, TerraMesh enables models to learn robust cross-modal relationships that are invariant to atmospheric conditions, a critical capability for all-weather monitoring [64]. These datasets collectively address the "sensor agnostic" challenge, where models must generalize across varying resolutions and physical measurement principles, as seen in frameworks that explicitly align representations across disparate sensors [107].

A critical dimension of these global datasets is the explicit modeling of seasonality and phenological cycles, which are essential for ecological and agricultural applications. While many early datasets sampled images based on calendar seasons, recent curation strategies have moved toward phenology-informed sampling to better capture vegetation dynamics. SSL4Eco introduces a global seasonal dataset specifically designed for ecology, utilizing a sampling strategy that aligns with local phenological cycles rather than fixed calendar dates, resulting in superior performance on biodiversity and vegetation tasks [91]. This focus on temporal density is also central to TESSERA, which generates pixel-level embeddings from global satellite time series, fusing optical and SAR data to create ready-to-use annual maps that preserve the temporal signal often lost in compositing [212]. Furthermore, the integration of hyperspectral data remains a frontier, with SpectralEarth providing a large-scale multi-temporal dataset from the EnMAP mission to pretrain models capable of handling the high dimensionality of hyperspectral imagery alongside traditional multispectral inputs [124].

The construction of these datasets also reflects a growing emphasis on standardization and accessibility within the pretraining ecosystem. Formats such as Zarr and cloud-optimized GeoTIFFs are becoming standard to handle the petabyte-scale volumes of data generated by missions like Copernicus, ensuring that data can be accessed efficiently without local download bottlenecks [229]. Comparative analyses reveal a clear trajectory from smaller, single-season datasets toward massive, multi-temporal, and globally covered resources like MajorTOM-Core and MMEarth, which incorporate digital elevation models and land use land cover data alongside optical and radar imagery [64].

Additionally, efforts like SeeFar aim to create satellite-agnostic datasets by combining multi-resolution commercial and public access images, enabling models to utilize historical data alongside higher-resolution imagery for greater flexibility during inference [73]. Despite this progress, challenges remain regarding the redundancy inherent in global satellite archives, which has spurred research into dynamic dataset curation and pruning strategies to maximize diversity and training efficiency [69]. The emergence of unified frameworks like EarthNets, which catalogs hundreds of datasets to standardize evaluation, further underscores the community's move toward a cohesive data infrastructure that supports the next generation of multi-sensor foundation models [272].

Domain-Specific, Regional, and Ecologically Biased Curation

The paradigm for constructing pre-training datasets in Geospatial Foundation Models (GFMs) has shifted from a singular pursuit of massive scale toward a nuanced strategy prioritizing geographic representativeness, ecological fidelity, and domain-specific diversity. While early hypotheses suggested that maximizing data volume would inherently yield superior generalization, empirical evidence now indicates that the spatial distribution and semantic composition of pre-training pools are critical determinants of downstream efficacy. A pivotal investigation into the spatial distribution of pre-training data reveals that balanced, globally representative compositions often outperform region-specific sampling strategies for general GFMs, highlighting that diversity and global coverage are paramount for robust feature learning [144]. However, this finding is not universal; the optimal sampling technique appears contingent upon the specific model architecture and the intended application domain, creating a nuanced tension between global generalizability and regional specialization that defines current curation efforts.

In ecological and environmental monitoring, the limitations of generic global datasets become particularly pronounced when addressing phenological cycles and biodiversity crises. Standard datasets frequently exhibit biases toward areas of high human activity or adhere to rigid calendar seasons, failing to capture local vegetative dynamics essential for accurate ecological modeling. To mitigate this, researchers have proposed phenology-informed sampling strategies that align data collection with local biological cycles rather than arbitrary dates. The introduction of SSL4Eco, a global seasonal dataset constructed using such strategies, demonstrates that capturing vegetation seasonality at a global scale significantly improves representation quality across diverse ecological tasks, outperforming models trained on calendar-based multi-date imagery [91]. This underscores the necessity of moving beyond simple temporal stacking to incorporate domain knowledge regarding biological timing into the curation process, ensuring that models learn the dynamics of life rather than just the static appearance of land cover.

Conversely, for highly specialized regions with unique spectral and structural characteristics, such as the Arctic, domain-specific curation proves superior to general-purpose pre-training. The Arctic presents distinct challenges, including diverse spectral signatures for semantically similar features and a scarcity of labeled data, which often confound models trained on temperate data. Studies focusing on pan-Arctic permafrost landform detection have shown that integrating geospatial location embeddings with Vision Transformers (ViTs) enhances model adaptation across these harsh environments [97]. Furthermore, a novel Arctic-focused remote sensing foundation model, developed through diversity-aware regional-scale image curation and masked autoencoder pre-training, achieved consistent improvements over both ImageNet-initialized baselines and general Earth observation foundation models like Prithvi-EO-2.0 [22]. By employing scalable affinity-propagation clustering to select representative training patches from vast archives, this approach successfully reduced the oversampling of visually repetitive or low-information areas, thereby preserving broad scene diversity within the specific domain. These results suggest that optimizing pre-training data distribution at a regional scale can produce more transferable representations for fine-scale mapping than general-purpose models trained on heterogeneous global data.

The challenge of data redundancy and heavy-tailed distributions is not limited to specific biomes but is a pervasive issue in Earth observation due to the repetitive nature of satellite imagery. Dynamic dataset curation strategies have been introduced to address this by iteratively pruning training sets to maximize diversity and balancedness without relying on pre-existing feature extractors. Such dynamic pruning has been shown to improve computational efficiency and feature quality, leading to better transferability in real-world applications, particularly in domains like ocean observation where data redundancy is high [69]. Similarly, in the context of crop type mapping, concerns regarding geospatial bias—where models trained on data-rich developed nations fail to generalize to data-scarce developing regions—highlight the need for careful consideration of geographic representation in pre-training pools [235].

Despite the advantages of regional specialization, there remains a strong argument for the utility of large-scale, multimodal datasets that span diverse geographic and sensor modalities to support broad applicability. Datasets like TerraMesh and SatlasPretrain aim to provide globally diverse, multimodal inputs that enable large-scale pre-training across optical, radar, and elevation data [64] [19]. These resources facilitate the development of models capable of handling varied input configurations, yet they must be carefully balanced to avoid the pitfalls of geographic bias. The interplay between global scale and local specificity suggests a hybrid approach may be necessary: leveraging massive, diverse datasets for foundational representation learning while employing targeted, domain-aware curation and

adaptation strategies-such as location embeddings or phenology-informed sampling-to refine models for specific ecological or regional tasks. This duality is further supported by findings that while global models provide strong baselines, domain-adapted pre-training on curated regional imagery yields significant performance gains for specialized applications, indicating that the "one-size-fits-all" paradigm is insufficient for the complex heterogeneity of Earth's surface [22] [91].

Temporal Dynamics, High-Frequency Data, and Irregular Time Series

The integration of high-temporal-resolution data into Earth Observation (EO) foundation models represents a paradigm shift from static image analysis to dynamic process modeling, yet this transition is fundamentally hindered by the inherent irregularity of Satellite Image Time Series (SITS). Traditional workflows often rely on annual or seasonal compositing to mitigate cloud cover and atmospheric noise, a practice that frequently discards vital phenological signals and short-term variability essential for real-time monitoring [212]. To overcome these limitations, recent research has pivoted toward architectures capable of ingesting raw, irregular, and high-frequency time series directly, acknowledging that effective modeling of fast-evolving phenomena requires preserving the continuous temporal signal rather than aggregating it into cloud-free mosaics [7]. The HighFM framework, for instance, adapts masked autoencoding strategies specifically for geostationary data, leveraging over 2 TB of SEVIRI imagery to capture the fine-grained temporal dynamics necessary for disaster detection and active fire tracking [7].

A primary challenge in processing SITS is the misalignment caused by varying revisit times and persistent cloud occlusion, which violates the assumption of regular temporal intervals made by standard convolutional or transformer architectures. The ALISE (ALigned Sits Encoder) framework addresses this by producing aligned latent representations from irregular and unaligned SITS, effectively handling spectral, spatial, and temporal dimensions simultaneously to support operational applications [156]. Similarly, TESSERA utilizes self-supervised learning to generate ready-to-use embeddings from pixel-level time series that are unevenly spaced and corrupted by noise, demonstrating that preserving the spectral-temporal signal yields superior performance in tasks ranging from crop type detection to fire scar mapping compared to methods that discard temporal fidelity [212]. Complementary to these encoder-based approaches, generative models like DiffusionSat and SAR-to-RGB translation techniques offer mechanisms to impute missing data or synthesize optical imagery from radar inputs, thereby filling gaps caused by cloud cover and enabling continuous monitoring even under adverse atmospheric conditions [187; 273].

Architectural innovations have further refined how temporal dependencies are captured, sparking a debate regarding the optimal factorization of space and time dimensions within pretraining objectives. While Vision Transformers (ViTs) have become prevalent, their quadratic computational scaling poses challenges for long, multitemporal sequences. SatMamba introduces State Space Models to achieve linear computational scaling, making it more efficient for processing long time series without sacrificing the ability to capture long-range dependencies [16]. In contrast, the Temporo-Spatial Vision Transformer (TSViT) argues for a temporal-then-spatial factorization strategy, positing that this ordering is more intuitive for SITS processing than the spatial-first approach common in natural image processing [98]. Furthermore, SatSwinMAE extends the SwinMAE architecture with hierarchical 3D masked autoencoding to effectively model multi-scale spatiotemporal dependencies, demonstrating significant improvements in land cover segmentation and flood mapping [21]. These diverse architectural choices highlight the ongoing effort to balance computational efficiency with the need to capture complex spatiotemporal relationships in foundation model pretraining.

The utility of these advanced temporal models is increasingly validated through their application in time-critical domains such as disaster response and agricultural monitoring, where the ability to process high-frequency data enables near real-time decision-making. This capability is further enhanced by edge computing frameworks that optimize onboard processing and data downlinking, addressing the bandwidth constraints of Low Earth Orbit satellites [225; 299]. For example, models adapted via Low-Rank Adaptation (LoRA) have shown robust generalization in wildfire burned-area mapping across diverse biomes and temporal shifts, proving that efficient fine-tuning of foundation models can support rapid deployment in emergency scenarios [240]. Additionally, the EarthPT model demonstrates the potential of autoregressive transformers to forecast future surface reflectances months in advance, offering a powerful tool for predicting vegetation evolution and land use changes [117].

Despite these advancements, contradictions and gaps remain regarding the standardization of temporal data handling and pretraining objectives within the broader data ecosystem. While some studies advocate for massive, global pre-training datasets like SSL4EO-S12 to learn generalizable temporal features, others emphasize the need for domain-specific sampling strategies, such as phenology-informed sampling in SSL4Eco, to better capture ecological seasonality that global datasets might miss [85; 91]. Moreover, the integration of causal relationships into pretraining objectives, as proposed by Causally Informed Variable-Step Forecasting (CI-VSF), suggests that current

self-supervised methods may not fully exploit the driver-response dynamics inherent in Earth systems [141]. The development of continuous spatiotemporal neural representations, such as LIANet, offers an alternative paradigm where models reconstruct imagery based solely on coordinates, potentially bypassing the need for discrete time-step alignment altogether [102]. As the field progresses, the synthesis of high-frequency data handling, irregular time series alignment, and efficient architectural designs will be paramount for realizing the full potential of foundation models in dynamic Earth monitoring.

Weak Supervision, Automated Labeling, and Synthetic Data Engines

The construction of large-scale training sets for Earth observation has traditionally been bottlenecked by the prohibitive cost and time required for manual annotation, prompting a paradigm shift toward weak supervision, automated labeling pipelines, and synthetic data generation. These approaches leverage existing geospatial products, coarse administrative statistics, and generative models to create robust datasets without relying on pixel-perfect human labels. A primary strategy involves utilizing free global land-cover products and geographic information systems (GIS) to derive weak labels for vast regions. For instance, methodologies have been developed to extend high-quality but geographically limited datasets, such as LANDFIRE's Existing Vegetation Type, to new regions like Canada by leveraging information-dense global representations [213]. Similarly, the integration of OpenStreetMap and other vector data sources provides explicit semantic relationships and geometry that can complement raster imagery, offering a pathway to unify raster perception with vector-based reasoning for more semantically grounded understanding [114].

Beyond leveraging existing maps, researchers are exploring techniques to learn from coarse label proportions rather than precise pixel annotations, a method particularly relevant for on-orbit training or scenarios where only aggregated statistics are available. By framing these challenges as Learning from Label Proportions (LLP) problems, it is possible to train lightweight models that achieve fine-grained spatial detail using only coarser tabular data, significantly reducing the dependency on detailed ground truth [209]. Furthermore, Multiple Instance Learning (MIL) approaches, such as Scene-to-Patch models, enable land cover classification using only high-level scene labels, bypassing the need for fully segmented datasets while still providing patch-level predictions [137]. This shift allows for the rapid development of new datasets for climate change mitigation and other critical applications where detailed annotation is infeasible.

Self-supervised learning (SSL) serves as a critical engine for automated labeling by learning representations from vast amounts of unlabeled data, effectively replacing or augmenting manual annotation. Various studies demonstrate that SSL can achieve reasonable accuracy with minimal fine-tuning, even without high-quality ground truth masks [269]. Contrastive representation learning has also proven effective, building generalizable representations from limited city samples that transfer well to unseen locations, although domain discrepancies between natural and abstract imagery remain a challenge [42]. To further enhance efficiency, dynamic dataset curation strategies have been proposed to prune redundant satellite imagery, ensuring that pre-training datasets are diverse and balanced, which improves feature quality for downstream tasks [69]. Additionally, soft contrastive learning frameworks that incorporate land-cover-generated multi-label supervision have been shown to resolve bottlenecks in contrastive learning, boosting the effectiveness of pre-training on multispectral and SAR imagery [248]. The efficacy of these methods is further supported by findings that pre-trained weights designed explicitly for Sentinel-2 outperform general pre-trained weights like ImageNet, especially in data-scarce regimes [235].

When labeled data is scarce or specific conditions are underrepresented, generative models act as powerful data engines to synthesize realistic satellite imagery. Diffusion models, tailored for remote sensing, can generate high-resolution samples conditioned on metadata such as geolocation, addressing the sparsity of text-based captions and enabling tasks like temporal generation and in-painting [187]. These generative capabilities are also being harnessed for few-shot object detection, where diffusion models synthesize diverse instance-level slices to augment training data for rare events, such as endangered species monitoring or disaster assessment [245]. The creation of automated data engines extends to object detection and segmentation, where systems like RemoteSAM utilize automatic data generation to create massive datasets comprising hundreds of thousands of image-text-mask triplets, covering diverse semantic categories that would be infeasible to annotate manually [78]. Similarly, the LAE-Label Engine automates the collection and annotation of remote sensing datasets to support open-vocabulary object detection, unifying multiple sources into a comprehensive training corpus [152].

Despite the promise of these automated and synthetic approaches, contradictions and challenges persist regarding the necessity of domain-specific pre-training versus general-purpose models. Some research suggests that self-supervised pre-training on remote sensing data offers only modest improvements over ImageNet pre-training in certain few-shot settings, questioning the cost-benefit ratio of curating massive domain-specific datasets [138]. Conversely, other studies argue that specialized foundation models significantly outperform general baselines, particularly when dealing with the unique spectral and spatial characteristics of satellite imagery, such as in SAR data analysis or multi-temporal tasks [105] [21]. The debate extends to the geographic distribution of pre-training data, with evidence suggesting that

balanced, globally representative data compositions often outperform region-specific sampling, highlighting the importance of diversity in automated dataset construction [144]. Ultimately, the synthesis of weak supervision, automated labeling, and generative data engines represents a paradigm shift, moving the field from labor-intensive annotation toward scalable, data-centric methodologies capable of addressing global monitoring challenges.

Benchmarking Frameworks, Evaluation Methodologies, and Standardization

The rapid proliferation of Geospatial Foundation Models (GFM) has necessitated the development of rigorous benchmarking frameworks to standardize comparisons across diverse architectures, tasks, and sensor modalities. Early efforts primarily focused on establishing baseline performance metrics for in-distribution tasks, yet recent literature increasingly critiques this approach, arguing that it fails to capture the complexities of real-world deployment. [3] introduced a foundational benchmark comprising six classification and six segmentation tasks to drive progress in Earth monitoring, providing a robust methodology for aggregating results across multiple baselines. Similarly, [119] addressed the scarcity of annotated data by creating a 400GB Sentinel-2 dataset with labels for building density, road segmentation, and land cover classification, enabling fair evaluation of models like Prithvi and SatMAE under varying shot regimes. While these frameworks established essential protocols for relative model comparison, they primarily measure performance under controlled, in-distribution conditions, which may not reflect operational realities.

A significant shift in the literature advocates for evaluating distributional robustness and generalization capabilities rather than mere accuracy on static test sets. [15] explicitly challenges the reliance on in-distribution metrics, introducing a testbed that pairs datasets across different time periods, geographies, and sensors to measure out-of-distribution performance. Their findings reveal that GFMs consistently perform 15-20% worse when faced with realistic distribution shifts, a degradation pattern similar to generic vision models, highlighting a critical gap in current model reliability. This concern is echoed by [38], which extends evaluation protocols to test generalization to satellites with no band overlap or additional bands, demonstrating that even state-of-the-art models suffer significant performance drops when encountering unseen sensor configurations. Furthermore, [56] expands the scope to open-world tasks, incorporating semantic and covariate shifts across five data domains to evaluate a model's ability to detect shifts and adapt without forgetting, a capability often overlooked in standard benchmarks. These studies collectively indicate that current leaderboards may overestimate model utility in dynamic environments.

Beyond robustness, the community is developing methodologies to assess model efficiency, scalability, and practical utility in resource-constrained environments. [143] provides empirical evidence that remote sensing models enter an overparameterized regime much earlier than their computer vision counterparts, retaining over 71% relative accuracy even when slimmed to 1% of their computational budget. This suggests that current scaling laws may not apply directly to Earth observation, prompting a re-evaluation of model size versus performance trade-offs. Complementing this, [90] proposes a cost-effective method to predict model performance on downstream tasks without exhaustive fine-tuning, utilizing "capabilities encoding" to simplify model selection. The push for standardization also extends to the accessibility of model outputs; [128] formalizes a taxonomy for pre-computed embedding products and extends TorchGeo with a unified API to decouple downstream analysis from model-specific engineering, thereby reducing barriers to reproducibility. Tools like [122] further support this by integrating domain-specific data modules and automated hyperparameter optimization to streamline the benchmarking process.

Operational deployment introduces further constraints that standard benchmarks often ignore, such as latency, compute limits, and specific application requirements. [104] argues that standardized evaluation tasks fail to capture operational variabilities like data heterogeneity and resource limitations, proposing a three-step protocol-requirements definition, adaptation strategy, and empirical testing-to bridge the gap between benchmark success and real-world utility. This perspective is supported by [57], which demonstrates that model compression and domain adaptation are critical for enabling onboard execution on resource-constrained space hardware, validating performance in actual flight environments rather than simulated ones. Additionally, [43] highlights the lack of dedicated data pipelines as a major obstacle, presenting an end-to-end framework that integrates automated data curation with task-specific distillation to produce compute-efficient models suitable for operational use. These works emphasize that future benchmarks must incorporate deployment feasibility as a core metric.

The evaluation landscape is also expanding to include specialized domains and multimodal capabilities. [202] introduces FoMo-Bench, a comprehensive benchmark covering 15 diverse datasets and multiple modalities specifically for forest monitoring, addressing the unique challenges of ecological applications. Similarly, [32] grounds benchmarking in the 17 Sustainable Development Goals, advocating for metrics that include energy efficiency and

ethical considerations alongside accuracy. In the realm of quantitative remote sensing, [67] and [34] emphasize the need to evaluate models on physically interpretable regression tasks rather than just classification, noting that traditional fidelity metrics often fail to correlate with downstream task utility. Finally, [194] addresses the limitations of existing benchmarks in handling ultra-high-resolution imagery, defining sub-tasks to evaluate the reasoning capabilities of multimodal large language models in complex remote sensing scenarios. Collectively, these works illustrate a maturing field moving from simple accuracy leaderboards toward holistic frameworks that prioritize robustness, efficiency, standardization, and real-world impact.

Robustness, Distribution Shift, and Open-World Evaluation

The transition from controlled benchmark environments to real-world deployment reveals a critical vulnerability in current Geospatial Foundation Models (GFM): a significant degradation in performance when faced with distribution shifts. While standard benchmarks often report impressive in-distribution accuracy, they frequently fail to capture the complexities of operational settings characterized by temporal, geographic, and sensor heterogeneity. To address this gap, recent literature has introduced specialized testbeds designed explicitly to measure robustness against realistic distribution shifts. A primary example is EarthShift, which serves as the first public testbed for benchmarking robustness across multiple realistic distribution shifts encountered in remote sensing, including changes in time periods, geographies, scales, and sensors [15]. Experiments conducted on EarthShift involving eight GFMs and eleven tasks demonstrate that these models consistently perform 15-20% worse in out-of-distribution (OOD) settings compared to in-distribution baselines, regardless of model architecture, size, or pre-training strategy [15]. This finding underscores a systemic issue where current foundation models possess robustness levels similar to generic vision models and fully supervised approaches, highlighting an urgent need to prioritize distributional robustness over pure accuracy metrics.

The challenge of distribution shift extends beyond temporal and geographic variations to include significant sensor heterogeneity. As the constellation of remote sensing satellites expands, models must generalize to new sensors with varying spectral bands, a capability often lacking in current architectures. GeoCrossBench addresses this by evaluating cross-band generalization, revealing that even state-of-the-art foundation models suffer a 2-4x drop in performance when generalizing to satellites with no band overlap with the training data [38]. Similarly, OpenEarthSensing provides a large-scale fine-grained benchmark for open-world remote sensing, encompassing diverse data domains with significant covariate shifts, including RGB satellite, aerial, multispectral, and infrared domains [56]. Evaluations on this benchmark confirm that existing methods struggle to detect semantic shifts or adapt to covariate shifts without forgetting previously learned knowledge, indicating that current models are not yet ready for the continuous influx of novel data characteristic of open-world scenarios [56]. Furthermore, the reliability of models in high-stakes scenarios is questioned by ExEBench, which benchmarks foundation models on extreme Earth events such as floods, wildfires, and heatwaves [52]. This benchmark highlights that models often inherit biases from training data that challenge their performance over extreme values, necessitating a shift in evaluation focus toward reliability under such critical conditions [52].

The root causes of these robustness failures are multifaceted, involving both data composition and architectural rigidity. Research into the spatial distribution of pre-training data suggests that balanced and globally representative data compositions often outperform region-specific sampling, yet many models remain biased toward data-rich regions, limiting their generalization capabilities in data-scarce areas like Sub-Saharan Africa [144]. This limitation is starkly illustrated in crop yield prediction studies, where frozen foundation model embeddings offered no meaningful advantage over traditional spectral features under a rigorous Leave-One-Country-Out evaluation, suggesting that the primary bottleneck is country-level yield distribution shift rather than feature representation deficiencies [89]. Additionally, architectural constraints play a role; many models are rigid regarding input patch sizes and sensor types. In contrast, compute-adaptive models like THOR demonstrate that unifying data from multiple sensors with flexible patch sizes can improve performance in data-limited regimes, suggesting that architectural flexibility is a key component of robustness [27]. Similarly, neuroplasticity-inspired architectures like DOFA utilize dynamic hypernetworks to process inputs from diverse sensors flexibly, achieving state-of-the-art performance across multiple downstream tasks and generalizing well to unseen modalities [13].

Operational deployment further exposes the disconnect between benchmark performance and real-world utility. Lessons from the WorldCereal global crop-mapping system indicate that standardized evaluation tasks often fail to capture real-world complexities such as data heterogeneity, resource constraints, and application-specific requirements [104]. A structured protocol for integrating foundation models into operational systems emphasizes the need for empirical testing that simulates real-world scenarios rather than relying solely on controlled benchmarks [104]. To mitigate performance drops in OOD settings, adaptation strategies such as Low-Rank Adaptation (LoRA) have shown promise. Studies on flood segmentation and wildfire mapping demonstrate that LoRA-based fine-tuning can significantly improve OOD performance while updating less than 1% of parameters, offering a computationally efficient pathway to enhance model robustness in time-critical scenarios [116] [240]. Moreover, the redundancy

inherent in remote sensing foundation models suggests that post-hoc slimmability can be leveraged to create resource-efficient variants that maintain high relative accuracy even under significant parameter reduction, providing a practical strategy for deployment in resource-constrained environments [143].

The consensus across these studies is that the field must pivot from maximizing in-distribution accuracy to ensuring reliability under shift. While benchmarks like GeoBench and PhilEO Bench have been instrumental in standardizing evaluations, they often assume static conditions that do not reflect the dynamic nature of Earth observation [3] [119]. The emergence of benchmarks like EarthShift, GeoCrossBench, and ExEBench represents a necessary evolution in evaluation methodologies, forcing the community to confront the limitations of current models in open-world settings. Future research must therefore focus on developing architectures that are inherently robust to sensor and domain shifts, curating pre-training datasets with global representativeness, and establishing evaluation protocols that prioritize operational reliability over theoretical peak performance.

Sensor-Agnostic, Multimodal, and Downstream-Integrated Metrics

The evolution of evaluation protocols in remote sensing has fundamentally shifted from assessing isolated visual fidelity to measuring generalization across heterogeneous sensor configurations and utility in complex downstream tasks. A primary driver of this transition is the recognition that traditional metrics often fail to capture the operational value of foundation models in real-world scenarios. Historically, super-resolution and reconstruction tasks have relied heavily on pixel-level fidelity metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM), yet these measures frequently exhibit weak or even negative correlations with performance in practical applications. [25] explicitly addresses this disconnect by introducing GeoSR-Bench, a benchmark that evaluates super-resolution models based on their impact on downstream tasks like land cover segmentation and biophysical variable estimation. The findings from this study reveal that improvements in traditional fidelity metrics can sometimes negatively correlate with task performance, underscoring the necessity of integrating downstream tasks directly into the evaluation pipeline rather than relying on abstract representation quality. This sentiment is reinforced by [30], which notes that pre-training benefits are highly task-dependent, particularly when the fine-tuning task diverges from the pre-training objective, suggesting that evaluation must be context-specific and aligned with the intended application.

Parallel to the move toward task-integrated metrics, there is a critical demand for evaluation frameworks that assess a model's ability to generalize across arbitrary sensor configurations without retraining. Traditional foundation models are often rigid, tailored to specific spectral bands or spatial resolutions, which limits their adaptability in the dynamic landscape of Earth observation. To address this, recent research has prioritized the development of sensor-agnostic architectures, necessitating new evaluation protocols that test performance on unseen modalities. [11] proposes treating co-located images from different sensors as natural augmentations, utilizing cross-channel attention to achieve state-of-the-art performance on diverse sensor configurations, thereby setting a new standard for evaluating cross-sensor generalization. Similarly, [13] introduces a wavelength-conditioned dynamic hypernetwork that emulates neuroplasticity, allowing a single model to dynamically adjust its weights based on the central wavelengths of input bands, thus generalizing effectively to unseen modalities. These approaches are complemented by [125], which employs spectrum-independent tokenizers and multi-scale adaptive patch embeddings to handle missing bands and varying resolutions, demonstrating robustness in cross-sensor settings. Furthermore, [27] advances this flexibility by unifying data from multiple Copernicus satellites with varying resolutions into a single model that supports dynamic trade-offs between computational cost and feature resolution at inference time, introducing "compute-adaptability" as a novel dimension for model assessment.

The assessment of multimodal capabilities has also expanded to include extremely large ultra-high-resolution imagery and complex reasoning tasks, pushing evaluation beyond simple classification or detection. [194] establishes a comprehensive benchmark for evaluating perception and reasoning in ultra-high-resolution scenarios, highlighting that existing models struggle with the complex semantic relationships inherent in such data. This focus on high-level cognition extends to [275], which evaluates models on their ability to not only understand spatiotemporal changes but also forecast future scenes, integrating geographic metadata to ensure physical plausibility. Additionally, [298] proposes a unified workflow evaluated on multi-granularity reasoning tasks, utilizing reinforcement learning to enable autonomous interpretation of user intent across object, region, and pixel levels. These benchmarks collectively push the field toward evaluating models that can reason about spatiotemporal dynamics and generate actionable insights rather than merely producing static predictions.

Efficiency and adaptability remain central themes in modern evaluation frameworks, particularly regarding model scaling and resource constraints, leading to the proposal of "slimmability" as a key metric for robustness. [143] challenges the prevailing scaling paradigms by demonstrating that remote sensing foundation models exhibit significant representational redundancy, allowing for substantial parameter reduction with minimal performance loss. This finding suggests that evaluation metrics should increasingly account for computational efficiency and the ability to maintain accuracy under reduced computational budgets as key indicators of model robustness. Moreover, [90] offers a cost-effective method for predicting model performance across multiple tasks without extensive fine-tuning, facilitating faster comparison and selection of foundation models. The integration of these diverse evaluation strategies—ranging from sensor-agnostic generalization and downstream task utility to ultra-high-resolution reasoning and efficiency

metrics-marks a maturation of the field, moving towards holistic frameworks that better reflect the complexities of real-world Earth observation applications.

Reproducibility, Standardization, and Embedding Products

The rapid proliferation of Geospatial Foundation Models (GFMs) has created a paradoxical landscape where powerful representational capabilities are increasingly accessible, yet their practical deployment and comparative evaluation remain hindered by significant fragmentation. A primary bottleneck identified across the literature is the lack of standardization in data formats, model interfaces, and evaluation protocols, which creates substantial engineering overhead for practitioners. To address this, recent initiatives propose treating pre-computed embeddings not merely as intermediate model outputs, but as first-class "geospatial products" that can be standardized, shared, and interoperably accessed. This shift aims to decouple downstream analysis from the complexities of model-specific engineering, thereby fostering reproducibility and accelerating the adoption of foundation models in operational settings [128]. By formalizing a taxonomy comprising Data, Tools, and Value layers, researchers argue that standardizing access to these embedding products through unified APIs is essential to overcome the current ecosystem of incompatible formats and resolutions [128].

Several technical frameworks have emerged to operationalize this vision of standardized access. The `rs-embed` library, for instance, introduces a region-of-interest-centric interface that allows users to retrieve embeddings from diverse models for any location and time range with minimal code, directly addressing the heterogeneity in model release formats and input specifications [203]. Similarly, extensions to existing libraries like `TorchGeo` provide unified mechanisms for loading and querying diverse embedding products, effectively treating them as standard geospatial datasets [128]. These tools are complemented by web-based applications such as `EarthEmbeddingExplorer`, which democratize access to global embedding datasets by enabling cross-modal retrieval through natural language and visual queries, thus bridging the gap between static research artifacts and dynamic analytical workflows [259]. Furthermore, the release of massive, analysis-ready embedding fields, such as those generated by AlphaEarth Foundations, demonstrates the viability of providing global, annual layers that can be utilized for mapping without the need for re-training or complex inference pipelines [49].

However, the transition from benchmark performance to real-world utility requires more than just standardized data access; it demands rigorous and context-aware evaluation methodologies. Standardized benchmarks like `GeoBench` and `PhilEO Bench` have been instrumental in establishing baseline comparisons across models, yet they often fail to capture the complexities of operational environments, such as data heterogeneity, resource constraints, and specific application requirements [104]. Critics note that while models may perform well on controlled benchmarks, they frequently suffer significant performance degradation when faced with out-of-distribution scenarios involving new geographies, sensors, or time periods, as highlighted by the `EarthShift` benchmark [15]. Consequently, there is a growing consensus that evaluation protocols must evolve to include metrics for robustness, transferability, and energy efficiency, moving beyond simple accuracy scores to ensure models are fit for purpose in diverse real-world applications [32].

To bridge the gap between benchmarking and deployment, structured protocols for operational integration are being proposed. These protocols emphasize the necessity of defining clear application requirements and hypotheses before model selection, adapting models to domain-specific data constraints, and conducting empirical testing under realistic conditions [104]. This approach contrasts with the common practice of vanilla fine-tuning, which often results in monolithic models that are computationally expensive and difficult to deploy. In response, frameworks like `InstaGeo` offer end-to-end solutions that integrate automated data curation, task-specific model distillation, and deployment tools, significantly reducing the time and computational resources required to move from raw data to operational insights [43]. Additionally, parameter-efficient adaptation techniques, such as Low-Rank Adaptation (LoRA) and embedding deflection strategies, are gaining traction as methods to tailor large foundation models to specific tasks without the prohibitive costs of full fine-tuning, further enhancing the feasibility of deploying these models in resource-constrained environments [240; 48].

Despite these advances, challenges regarding data diversity and model generalization persist. Studies indicate that the geographic distribution of pre-training data significantly impacts model performance, with globally representative datasets often yielding better generalization than region-specific sampling [144]. Moreover, the inability of current models to effectively handle cross-sensor generalization remains a critical limitation, prompting the development of "any-sensor" models and benchmarks like `GeoCrossBench` to evaluate performance across varying spectral bands and resolutions [38; 11]. The community is also increasingly recognizing the need for open-source governance and

standardized software design practices to ensure the long-term sustainability and reproducibility of geospatial machine learning libraries [127]. Ultimately, achieving true reproducibility and standardization requires a concerted effort to align data sharing practices, code release policies, and evaluation metrics, ensuring that foundation models serve as reliable and accessible tools for addressing global Earth observation challenges.

Multimodal Fusion, Cross-Modal Learning, and Vision-Language Integration

The integration of heterogeneous data sources represents a pivotal shift in Earth observation, moving from rigid, single-sensor analysis toward comprehensive multimodal understanding capable of handling arbitrary input configurations. Early fusion approaches often struggled with the diverse spectral and spatial characteristics of real-world data, necessitating architectures that treat sensor variability as a feature rather than a constraint. To address this inflexibility, recent foundation models have introduced dynamic mechanisms inspired by biological neural plasticity. For instance, the Dynamic One-For-All (DOFA) model utilizes a wavelength-conditioned dynamic hypernetwork to generate network weights based on the central wavelengths of input spectral bands, enabling a unified Transformer framework to adaptively process data from distinct satellite sensors, including those unseen during pre-training [13]. Similarly, Panopticon advances sensor-agnostic generalization by treating co-located multi-sensor images as natural augmentations and employing cross-attention over channels to flexibly embed arbitrary combinations of optical and Synthetic Aperture Radar (SAR) bands [11]. Expanding on this adaptability, THOR introduces a "compute-adaptive" architecture that unifies data from Copernicus Sentinel-1, -2, and -3 satellites, allowing a single set of pre-trained weights to be deployed at inference with any patch size to dynamically trade off between computational cost and feature resolution [27].

Contrasting with dynamic weight generation, other strategies focus on cross-modal alignment and contrastive learning to bridge the gap between disparate data types such as optical, radar, and LiDAR. The DUNIA framework learns pixel-sized embeddings by aligning images with full-waveform LiDAR data in a contrastive manner, facilitating zero-shot applications in canopy height mapping and land cover classification that outperform specialized supervised models in low-data regimes [216]. In the realm of optical and SAR integration, CLOSP utilizes text as a semantic bridge to align unpaired optical and SAR images into a unified embedding space, significantly improving text-to-image retrieval performance and transferring semantic knowledge from the optical domain to the more challenging SAR modality [175]. Furthermore, X-STARS introduces a Multi-Sensor Alignment Dense loss to align representations across sensors with vastly different resolutions, enabling continual pre-training that adapts low-resolution models to high-resolution sensors without requiring massive labeled datasets [107]. These methods highlight a divergence in fusion strategies: while some models like MAESTRO optimize masked autoencoding with spectral priors for multitemporal and multispectral data [95], others like MultiMAE adopt a multi-task pre-training strategy that reconstructs diverse input modalities including spectral, elevation, and segmentation data to ensure robust transfer learning [296].

The convergence of vision and language modalities has further expanded Earth observation capabilities, enabling complex querying, reasoning, and open-vocabulary understanding. Large Vision-Language Models (VLMs) are increasingly being adapted to interpret multi-sensory observations through interactive dialogues. EarthDial transforms complex Earth observations into natural language interactions, supporting tasks ranging from visual grounding to change detection across RGB, SAR, and multispectral modalities by leveraging an extensive instruction tuning dataset comprising over 11 million pairs [215]. However, standard VLMs often struggle with the ultra-high resolution and unique semantic relationships inherent in remote sensing imagery. Benchmarks like XLRs-Bench have revealed significant gaps in current models' ability to handle real-world scenarios, showing that even high-capacity models fail to adequately perceive and reason over extremely large images [194]. Addressing these limitations, specialized models like LISAT are designed to reason over complex user queries to perform segmentation, outperforming general open-domain models on geospatial reasoning tasks by integrating domain-specific pre-training [71].

Advanced reasoning capabilities are also being integrated into unified workflows that go beyond simple description to enable autonomous inference. RemoteReasoner proposes a unified geospatial reasoning workflow that employs reinforcement learning to enable autonomous inference paths for multi-granularity tasks, allowing the model to interpret nuanced user intents without task-specific fine-tuning [298]. Similarly, RS-WorldModel unifies spatiotemporal change understanding with text-guided future scene forecasting, leveraging geographic metadata to predict plausible futures and explain observed changes within a single framework [275]. The integration of Large Language Models (LLMs) also extends to agent-based systems; Earth AI orchestrates a family of foundation models across imagery, population, and environment domains using a Gemini-powered reasoning agent to solve complex,

multi-step crisis response queries [169]. Despite these advancements, challenges remain regarding the computational demands of training such massive multimodal systems and the scarcity of high-quality annotated datasets for instruction tuning [18]. Moreover, while models like EarthDial and RemoteSAM demonstrate strong performance, the field continues to grapple with the need for standardized benchmarks that adequately evaluate open-vocabulary detection and reasoning across diverse sensor modalities and geographic regions [152][78]. The evolution from early fusion techniques to sophisticated cross-attention and agentic reasoning marks a critical transition toward generalist AI systems capable of holistic Earth system analysis.

Optical, SAR, and Hyperspectral Fusion Techniques

The integration of optical, Synthetic Aperture Radar (SAR), and hyperspectral imagery represents a critical frontier in Earth observation, aiming to synthesize the spectral richness of hyperspectral data, the interpretability of optical sensors, and the all-weather structural sensitivity of SAR. A primary challenge in this domain is the significant domain gap arising from differing physical properties, spatial resolutions, and spectral alignments across sensors. Recent advancements have moved beyond simple concatenation toward sophisticated fusion architectures that leverage cross-modal attention, shared backbones, and sensor-parameter-aware embeddings to create robust, sensor-agnostic representations. A prevailing strategy involves the development of foundation models capable of processing heterogeneous inputs through unified encoding mechanisms, addressing the underrepresentation of specific modalities like hyperspectral imagery in multimodal pretraining [72]. This architecture utilizes a hierarchical transformer that combines spectral tokenization for hyperspectral inputs with sensor-specific encoders for optical and SAR data, effectively bridging the gap between high-dimensional hyperspectral cubes and lower-channel observations. Similarly, [11] proposes an any-sensor framework built on the DINOv2 architecture, which treats images of the same geolocation across different sensors as natural augmentations. By incorporating cross-attention over channels and encoding wavelength and sensing modes, Panopticon achieves state-of-the-art performance on diverse sensor configurations, demonstrating that flexible patch embedding mechanisms can effectively handle arbitrary combinations of optical and SAR channels without retraining.

Addressing the specific misalignments in spectral and spatial resolutions, several studies emphasize the importance of embedding sensor physics directly into the model architecture rather than treating multi-sensor data as generic image tensors. [191] introduces a versatile sensor parameter encoding module that injects ground-sampling distance and spectral response functions into image embeddings. This approach allows the model to differentiate between sensors and understand the correlation between sensor parameters and the observed signal, facilitating effective fusion even when spectral bands do not perfectly match. Complementing this, [125] employs a spectrum-independent tokenizer and multi-scale adaptive patch embeddings to accommodate arbitrary band compositions and resolution scales, ensuring that distinctive spectral characteristics are preserved even when bands are missing or newly introduced. These methods collectively suggest that explicit modeling of sensor parameters is superior to treating multi-sensor data as generic image tensors, enabling models to generalize across sensor generations and missions.

The fusion of optical and SAR data specifically benefits from techniques that align their disparate feature spaces, often using text or geographic context as a bridging modality. [175] presents CLOSP, a framework that uses text as a bridge to align unpaired optical and SAR images into a unified embedding space. By transferring rich semantic knowledge from the optical domain to SAR through indirect interaction, this method overcomes the inherent difficulty of interpreting SAR imagery. Furthermore, [Multi-Modal, Multi-Scale Representation Learning for Satellite Imagery Analysis Just Needs a Good Alibi] introduces Scale-ALiBi, a linear bias transformer attention mechanism that encodes relationships between image patches at different ground sample distances. This allows for the native processing of multi-modal and multi-resolution imagery, effectively fusing low-resolution SAR and optical data with high-resolution optical inputs through a triple-contrastive architecture. However, the transferability of such fusion models is not guaranteed; [184] observes that the ranking of optical models on optical change detection does not transfer one-to-one to SAR amplitude change detection, indicating strong sensitivity to preprocessing alignment with pretraining statistics. This suggests that fusion models trained on specific sensor pairs may struggle with generalization if the physical characteristics of the input data shift.

In the realm of hyperspectral and multispectral fusion, self-supervised modality decoupling has emerged as a powerful paradigm to reduce redundancy and aggregate concise representations. [266] proposes MossFuse, a framework that decouples shared and complementary information across modalities using subspace clustering loss, outperforming direct fusion methods that lack effective supervision. Similarly, [291] utilizes continuous low-rank factorization to capture spatial and spectral information via neural representations, enabling fusion at user-desired resolutions without retraining. These techniques highlight a shift from supervised fusion reliant on paired high-resolution ground truth to self-supervised strategies that exploit the intrinsic correlations between low-resolution hyperspectral and high-resolution multispectral data. Despite these advancements, contradictions remain regarding the efficiency of these complex architectures. [143] reveals that remote sensing foundation models operate in a fundamentally different

representational regime than computer vision models, exhibiting high redundancy that allows for significant parameter reduction without performance loss. This finding implies that current fusion architectures may be overparameterized, and future efforts should focus on efficient, slimmable designs that maintain cross-modal fidelity while reducing computational overhead. The synthesis of these diverse approaches underscores the necessity of architectures that are not only multimodal but also physically aware, computationally efficient, and adaptable to the irregular and unaligned nature of real-world satellite time series.

Vision-Language Models and Open-Vocabulary Understanding

The integration of Large Language Models (LLMs) with vision encoders has fundamentally transformed remote sensing analysis, shifting the paradigm from closed-set classification to open-vocabulary understanding where models interpret satellite imagery through natural language queries. Traditional approaches often relied on disjointed visual feature extractors and language processors, limiting their capacity to handle open-ended questions or unseen classes without extensive retraining [130]. To overcome these limitations, recent advancements have focused on generative Large Vision-Language Models (LVLMs) that condition natural language generation on both visual and textual inputs, enabling fluent and accurate responses to complex geospatial inquiries. However, a significant domain gap exists between natural images and satellite data, characterized by varying spectral bands, diverse spatial resolutions, and complex geospatial structures that general-purpose models fail to interpret effectively [130].

Addressing this domain gap has sparked a debate regarding the necessity of massive domain-specific pretraining versus efficient adaptation strategies. While some studies argue that billion-scale foundation models trained specifically on remote sensing data achieve superior performance and data efficiency, suggesting that scaling model parameters and domain-specific data is crucial [112], other research challenges this assumption. Systematic studies indicate that self-supervised pretraining on satellite imagery offers only modest improvements over ImageNet pretraining in few-shot settings, questioning the cost-effectiveness of massive domain-specific pretraining efforts [138] [29]. In response to these conflicting findings, innovative alignment strategies have emerged that bypass the need for textual annotations entirely. One notable approach utilizes co-located ground-level internet imagery as an intermediary to align satellite image encoders with general vision-language models like CLIP, enabling zero-shot, open-vocabulary tasks such as classification and segmentation with performance gains exceeding supervised baselines [249]. Furthermore, comprehensive metadatasets like SATIN have been developed to evaluate zero-shot transfer capabilities across diverse resolutions and geographic areas, revealing that even state-of-the-art models struggle with the heterogeneity of remote sensing data, thus highlighting the need for specialized adaptation strategies [106].

Beyond simple classification and question answering, the field has expanded into complex reasoning and instruction-oriented segmentation. Models like LISAT are designed to reason over complex user queries to identify and segment objects within satellite scenes, significantly outperforming existing geospatial foundation models on reasoning segmentation tasks by leveraging curated geospatial reasoning datasets [71]. This evolution extends to training-free frameworks that interpret implicit instructions for object counting, detection, and segmentation without relying on confidence thresholds, demonstrating efficiency in handling open-ended and open-subclass scenarios [283]. The complexity of remote sensing imagery, particularly ultra-high-resolution (UHR) data, necessitates innovative architectural solutions to preserve spatial context while managing computational costs, as standard VLMs often fail when resizing UHR images, leading to significant information loss [194]. Frameworks such as ImageRAG address this by transforming image analysis into a context selection task, selectively retrieving relevant portions of gigapixel images to answer queries accurately [231]. Similarly, hierarchical approaches like Aquila introduce spatial feature integration modules to support high-resolution inputs and achieve deep visual-language alignment, significantly improving the representation of complex visual information [241].

The evolution of these models also extends to agent-based systems that automate task planning and unified reasoning workflows. Remote Sensing ChatGPT utilizes an LLM-powered agent to connect various AI models, performing task planning and execution iteratively to solve complex interpretation tasks based on user requests [131]. Such systems represent a move towards unified workflows where a single model can handle multi-granularity reasoning tasks, from object-level detection to pixel-level segmentation, through autonomous reasoning and reinforcement learning [298]. Additionally, the expansion into multi-sensory observations is evident in models like EarthDial, which supports multispectral, SAR, and multi-temporal imagery through extensive instruction tuning, enabling interactive dialogues for diverse Earth observation tasks [215]. Despite these advancements, contradictions remain regarding the optimal training paradigm, with some approaches favoring parameter-efficient fine-tuning techniques like LoRA to adapt general models efficiently [180], while others advocate for training-free frameworks that leverage the inherent reasoning capabilities of existing large models [189]. As the field progresses, the focus remains on enhancing generalization across unseen classes and domains, ensuring that VLMs can robustly interpret the vast and diverse data streams of

modern Earth observation systems while addressing the unique challenges of scale, resolution, and spectral diversity inherent to satellite imagery.

Handling Missing Modalities, Channels, and Cross-Modal Alignment

The inherent heterogeneity of Earth observation (EO) data, characterized by varying spectral bands, spatial resolutions, and sensing modalities, necessitates foundation models that are robust to incomplete or irregular inputs, a common occurrence in operational scenarios due to sensor failures, cloud cover, or the integration of new satellite missions. Traditional architectures often fail when faced with missing channels or unseen sensor configurations, prompting a shift toward strategies that treat sensor diversity as a feature rather than a constraint. A primary approach involves explicitly encoding physical sensor parameters into the model's embedding space to preserve spectral identity even when bands are missing or newly introduced. For instance, [125] introduces a spectrum-independent tokenizer that assigns dedicated band embeddings to each channel, enabling the model to handle arbitrary band compositions and resolution scales without retraining. Similarly, [191] encodes specific sensor parameters, such as ground sampling distance and spectral response functions, directly into image embeddings, facilitating cross-sensor training and sensor-independent inference by making the model physically aware of the observing instrument.

Complementing these static adaptations, dynamic hypernetworks inspired by neural plasticity offer a flexible solution for adapting to unseen modalities through weight generation rather than fixed embeddings. [13] proposes the Dynamic One-For-All (DOFA) model, which utilizes a wavelength-conditioned dynamic hypernetwork to generate network weights based on the central wavelengths of input spectral bands, allowing a single unified Transformer backbone to adapt its internal structure to diverse and previously unseen sensor configurations. This mechanism effectively emulates the brain's capacity to reorganize in response to novel stimuli, enabling the model to process inputs from distinct satellite sensors flexibly without requiring separate encoders for each modality. In a similar vein, [11] extends the DINOv2 framework by treating co-located images from different sensors as natural augmentations and incorporating cross-attention over channels, allowing the model to process arbitrary combinations of optical and synthetic aperture radar (SAR) data by encoding wavelength and mode information. These methods collectively demonstrate that explicitly modeling the physical properties of sensors or dynamically adjusting network weights is crucial for robustness against missing or varying channels, moving the field away from rigid, sensor-specific models toward open-world foundation models.

Ensuring alignment across different modalities and temporal instances is equally critical for leveraging multi-source data, particularly when resolutions and geometries differ significantly. Cross-modal alignment techniques enable models to learn shared representations despite these discrepancies, often by aligning satellite imagery with complementary data sources. [216] addresses the gap between coarse patch-sized embeddings and fine-grained pixel-level tasks by aligning satellite imagery with full-waveform LiDAR data through contrastive learning, producing pixel-sized embeddings that facilitate zero-shot applications across diverse environmental monitoring tasks. Furthermore, [107] introduces the Multi-Sensor Alignment Dense (MSAD) loss to align representations across sensors with vastly different resolutions, such as low-resolution Sentinel-2 and high-resolution aerial data, enabling continual pretraining frameworks where models adapted to low-resolution data can be effectively transferred to high-resolution sensors. The importance of alignment extends to the temporal domain, where [156] proposes the ALigned Sits Encoder (ALISE), which produces aligned latent representations from irregular and unaligned satellite image time series, preserving spatial resolution while accounting for temporal inconsistencies common in operational applications.

Masked image modeling (MIM) has also been adapted to enhance resilience against missing data by training models to reconstruct masked portions of the input using available context, thereby learning deep inter-channel dependencies. [296] employs a multi-modal, multi-task masked autoencoder that reconstructs diverse input modalities, including spectral, elevation, and segmentation data, ensuring the model learns robust features capable of transferring to downstream tasks even when the input modality structure differs from the pre-training configuration. Additionally, [80] enhances MIM by incorporating domain-specific spectral indices to guide dynamic token masking toward informative regions, improving reconstruction capabilities even with high mask ratios. These reconstruction-based objectives force the model to infer missing information from the remaining context, significantly improving performance in data-sparse environments.

The flexibility afforded by these architectural innovations also supports zero-shot and few-shot learning capabilities, reducing the dependency on labeled data for new sensor configurations. [176] demonstrates a training-free approach that adapts non-RGB inputs to the visual space of standard RGB-only large multi-modal models via guided instructions

and chain-of-thought reasoning, leveraging the reasoning capabilities of generalist models to interpret specialized multi-spectral data without extensive retraining. Moreover, [14] shows that embeddings produced by models trained on multiple modalities can support downstream tasks with significantly reduced labeled data requirements, even in contexts with missing modalities or channels. The convergence of modality-aware masking, dynamic embeddings, and cross-modal alignment techniques signifies a paradigm shift toward flexible foundation models capable of adapting to the dynamic and heterogeneous nature of real-world EO data.

Reasoning, Scene Graphs, and Agentic Workflows

The frontier of multimodal integration in Earth observation is undergoing a fundamental paradigm shift from passive perception to active cognitive reasoning, where models are increasingly expected to generate structured insights and orchestrate complex workflows rather than simply classifying pixels. This transition is most visibly manifested in the move toward scene graph generation (SGG), a task that requires models to understand not just the presence of objects but the intricate semantic and spatial relationships between them. While SGG has matured in natural imagery, its application to large-size satellite imagery presents unique challenges due to extreme scale variations and the necessity for long-range contextual dependencies that standard models often miss. To address these specific geospatial hurdles, recent work has introduced large-scale datasets like STAR and ReCon1M, which provide the dense annotations required to train models to extract subject-relationship-object triplets in complex remote sensing contexts [86] [92]. These data resources are complemented by novel architectural frameworks such as the context-aware cascade cognition (CAC) approach, which is explicitly designed to handle the holistic understanding required for very-high-resolution satellite data by cascading object detection, pair pruning, and relationship prediction [86].

Beyond static scene understanding, the field is increasingly adopting "world models" capable of spatial extrapolation and future state forecasting, moving beyond pixel-level synthesis to systems that internalize environment dynamics. Traditional generative models often lack the reasoning capacity to predict plausible futures based on physical constraints, a gap that the Remote Sensing-Oriented World Model framework attempts to bridge by formulating world modeling as direction-conditioned spatial extrapolation validated through the RSWISE benchmark [135]. Similarly, RS-WorldModel unifies spatiotemporal change understanding with text-guided future scene forecasting, leveraging a three-stage training paradigm that includes geo-aware generative pre-training and verifiable reinforcement optimization to ensure physical plausibility in predicted states [275]. These approaches signify a critical departure toward systems that can anticipate future scenarios, a capability vital for applications in disaster response and urban planning where interpreting current states is insufficient without forecasting their evolution.

The culmination of these reasoning and generative capabilities is found in agentic workflows, where large language models (LLMs) act as central orchestrators to solve multi-step crisis scenarios by decomposing complex queries into executable subtasks. The concept of "Earth AI" exemplifies this by integrating foundation models across imagery, population, and environment domains, coordinated by a reasoning agent that can synthesize actionable insights from heterogeneous data sources [169]. This agentic paradigm is further realized in systems like Remote Sensing ChatGPT, which utilizes an LLM to plan tasks, execute subtasks iteratively using specialized visual models, and generate coherent responses, thereby democratizing access to remote sensing analysis for non-experts [131]. In resource-constrained environments, such as low Earth orbit satellite networks, synergistic satellite-ground systems like SpaceVerse deploy compact models on-board for lightweight reasoning while offloading intensive tasks to ground stations, optimizing the trade-off between autonomy and computational power [17].

However, the path to robust agentic intelligence faces significant challenges regarding model autonomy and generalization, particularly concerning the reliance on supervised fine-tuning with annotated reasoning processes. Many existing frameworks are limited in their ability to generalize to unseen tasks or out-of-distribution categories because they are confined to predefined ground-truth sequences. In contrast, the RemoteReasoner framework proposes a unified workflow trained via reinforcement learning, enabling the model to autonomously explore inference paths and handle multi-granularity tasks without task-specific decoders [298]. This highlights a growing consensus that while supervised methods provide stability, reinforcement learning may be essential for developing the self-directed thinking required for open-ended geospatial problems. Additionally, the heterogeneity of input data remains a significant hurdle; frameworks like EarthDial and PyViT-FUSE attempt to unify multi-sensor inputs, including SAR and multispectral data, into interactive dialogues or single representations, yet the integration of these diverse modalities into a cohesive reasoning engine continues to be an active area of research [215] [118]. Ultimately, the evolution from perception to agency requires not only larger datasets and more powerful backbones but also novel training paradigms that prioritize causal reasoning, physical consistency, and the ability to operate effectively across the vast spectrum of Earth observation modalities [141] [262].

Adaptation Mechanisms, Transfer Learning, and Parameter Efficiency

The deployment of Earth Observation (EO) foundation models in heterogeneous environments necessitates robust adaptation mechanisms that bridge the gap between massive pre-training datasets and specific, often resource-constrained, downstream applications. A primary challenge lies in the architectural rigidity of current models, which often struggle with varying sensor modalities, spatial resolutions, and spectral band configurations. To address input heterogeneity, recent architectures have moved towards unified, sensor-agnostic frameworks. For instance, the Dynamic One-For-All (DOFA) model employs a wavelength-conditioned dynamic hypernetwork inspired by neural plasticity, allowing a single backbone to adapt its weights dynamically to unseen sensor modalities without retraining [13]. Similarly, Panopticon treats co-located multi-sensor images as natural augmentations and utilizes cross-attention over channels to process arbitrary combinations of optical and SAR data, effectively generalizing to future satellite missions [11]. The THOR model further advances this by enabling compute-adaptive inference, where a single set of pre-trained weights can operate at variable patch sizes and resolutions, facilitating a dynamic trade-off between computational cost and feature resolution [27]. These approaches collectively suggest that future foundation models must inherently possess the flexibility to handle the diverse and evolving landscape of EO sensors.

Beyond architectural adaptability, the efficiency of transferring knowledge to downstream tasks is critical, particularly when labeled data is scarce or computational resources are limited. Parameter-Efficient Fine-Tuning (PEFT) has emerged as a superior alternative to full fine-tuning, offering significant reductions in trainable parameters while maintaining or improving performance. Low-Rank Adaptation (LoRA) has demonstrated particular efficacy in EO contexts; for example, in flood segmentation tasks, LoRA-based fine-tuning outperformed full fine-tuning which was computationally infeasible, while also enhancing out-of-distribution generalization [116]. Similar gains have been observed in wildfire mapping, where LoRA updated less than 1% of parameters yet achieved the strongest cross-domain generalization across diverse biomes compared to decoder-only or full fine-tuning strategies [240]. Innovations in PEFT continue to evolve, with methods like KronA+ extending Kronecker product-based adaptation for hyperspectral image classification, utilizing merely 0.056% of trainable parameters to achieve competitive performance against state-of-the-art backbones [59]. Furthermore, the DEFLECT strategy introduces embedding deflection to adapt RGB-pretrained models to multispectral imagery with minimal overhead, leveraging spatial priors while enhancing spectral representation [48].

The necessity for such efficiency is underscored by findings that remote sensing foundation models exhibit high representational redundancy, allowing them to be "slimmed" post-hoc with significantly less performance degradation than their computer vision counterparts [143]. This characteristic enables the deployment of compact variants on resource-constrained space hardware, as demonstrated by reliable on-orbit inference aboard the International Space Station using compressed Vision Transformer-based models [57]. Knowledge distillation (KD) serves as another pivotal mechanism for reducing model complexity. Dynamic weighting KD frameworks have enabled lightweight student models to achieve over 90% accuracy in image classification while drastically reducing parameters and power consumption compared to their teacher models [88]. The InstaGeo framework further integrates task-specific distillation into an end-to-end workflow, producing student models up to 8 times smaller than vanilla fine-tuned teachers with negligible performance degradation, thereby accelerating the data-to-deployment cycle for operational monitoring [43].

Despite these advancements, contradictions and limitations persist regarding the generalizability of adapted models. While some studies highlight the superior transferability of foundation models in label-scarce regimes, others reveal significant performance drops when models encounter out-of-distribution shifts in geography, time, or sensor type. The EarthShift benchmark indicates that geospatial foundation models consistently perform 15-20% worse in out-of-distribution scenarios, suggesting that current adaptation strategies may not fully resolve robustness issues [15]. Similarly, evaluations in agricultural contexts have shown that while foundation model embeddings are competitive, they can suffer from limited spatial transferability and time sensitivity compared to purpose-built remote sensing models [146]. Moreover, the assumption that domain-specific pre-training always yields better representations is challenged by findings where generalist vision models outperform EO-specific models in cross-scene retrieval tasks, indicating that EO pre-training alone does not guarantee robustness to distribution shifts [63]. These discrepancies underscore the need for adaptation protocols that rigorously test models under operational variability and data

heterogeneity, as proposed in structured deployment frameworks that prioritize empirical testing alongside requirement definition [104]. Ultimately, the path forward involves balancing the efficiency of PEFT and distillation with the development of architectures and training strategies that explicitly target robustness across the diverse and dynamic conditions inherent to Earth observation.

Parameter-Efficient Fine-Tuning (PEFT) and Distillation

The operational deployment of massive Earth Observation (EO) foundation models is frequently bottlenecked by the prohibitive memory and computational costs associated with full fine-tuning, particularly in resource-constrained environments such as on-board satellite processing or rapid disaster response. To mitigate these barriers, Parameter-Efficient Fine-Tuning (PEFT) and knowledge distillation have emerged as indispensable adaptation mechanisms that decouple model capacity from trainable parameter count. Low-Rank Adaptation (LoRA) has established itself as a leading PEFT strategy, demonstrating that updating only a low-rank decomposition of the weight matrices can match or even exceed the performance of full fine-tuning while updating a negligible fraction of parameters. In flood segmentation tasks, LoRA-based adaptation of foundation models improved F1 scores by 6.66 points compared to frozen encoder baselines and proved computationally feasible where full fine-tuning was not, highlighting its utility in time-critical scenarios [116]. Crucially, the benefits of LoRA extend beyond mere efficiency; evidence suggests it enhances generalization to out-of-distribution (OOD) data. When tested on geographically distinct flood events, LoRA configurations demonstrated superior OOD performance compared to both frozen and fully fine-tuned counterparts, implying that restricting parameter updates may act as a regularizer against overfitting to source domain artifacts [116].

This generalization advantage is corroborated in large-scale wildfire mapping across diverse biomes, where LoRA adaptation of geospatial foundation models yielded the strongest cross-domain performance while updating less than 1% of parameters, outperforming both decoder-only and full fine-tuning approaches [240]. The efficacy of PEFT is not limited to optical imagery; it has been successfully extended to the complex spectral domains of hyperspectral imaging. Innovations such as KronA+, which applies distinct learning rates to Kronecker product-based adapter matrices, have achieved performance comparable to full fine-tuning for hyperspectral classification while utilizing merely 0.056% of trainable parameters and adding negligible storage overhead [59]. Furthermore, novel strategies like DEFLECT introduce stronger inductive biases by deflecting embeddings rather than adding weights, achieving on-par accuracy with significantly fewer parameters for multispectral adaptation [48]. These findings collectively challenge the assumption that exhaustive parameter updates are necessary for high performance, suggesting instead that the intrinsic rank of task-specific updates in EO is low.

Complementing PEFT, knowledge distillation (KD) offers a pathway to compress large teacher models into lightweight student architectures suitable for extreme edge deployment. Dynamic weighting mechanisms in KD frameworks allow student models to prioritize knowledge from teachers based on confidence levels, enabling compact architectures like ResNet8 to achieve over 90% accuracy with drastic reductions in parameters, FLOPs, and power consumption [88]. This compression is vital for on-orbit applications, where hardware limitations necessitate compact variants of Vision Transformers that maintain downstream task performance, as demonstrated by successful on-orbit inference aboard the International Space Station [57]. Distillation has also proven effective in complex fusion tasks; for instance, student models trained via cross self-attention mechanisms achieved superior super-resolution performance in hyperspectral and multispectral fusion with significantly reduced model sizes [258]. End-to-end frameworks like InstaGeo have further operationalized this by combining automated data curation with task-specific distillation, producing models up to eight times smaller than those from standard fine-tuning with minimal performance degradation across tasks like flood mapping and crop segmentation [43].

The success of these compression and adaptation techniques is underpinned by a fundamental property of remote sensing foundation models: inherent redundancy. Unlike computer vision models trained on natural images, which suffer sharp performance drops when slimmed, remote sensing models retain over 71% relative accuracy even when computational budgets are reduced to 1%, suggesting they operate in an overparameterized regime where task-relevant information is distributed redundantly [143]. This "slimmability" validates the use of PEFT and distillation not just as cost-saving measures, but as strategies that align with the representational nature of geospatial data. However, the literature reveals nuanced trade-offs; while PEFT excels in data-scarce regimes and domain shifts, some studies indicate that for tasks closely resembling the pre-training objective, such as reconstruction, full fine-tuning or even training from scratch with optimized hyperparameters can be equally effective [30]. Nevertheless, in practical applications involving limited ground-truth labels, such as intra-field crop yield regression, fine-tuning foundation models via parameter-efficient adapters consistently outperforms training architectures from scratch, confirming the value of

leveraging pretrained representations even under strict parameter budgets [297]. Collectively, these approaches transform massive, static foundation models into adaptable, efficient tools capable of deployment across the spectrum of Earth observation challenges, from cloud-based analysis to on-satellite inference.

Cross-Sensor, Cross-Band, and Domain Generalization

The rapid proliferation of Earth observation satellites has created a heterogeneous data landscape characterized by diverse spectral bands, spatial resolutions, and sensing modalities, presenting a significant challenge for model generalization. A primary obstacle in leveraging this data is the inability of models trained on specific sensors to generalize effectively to new satellites or geographic regions, a limitation starkly quantified by recent benchmarking efforts. The GeoCrossBench: Cross-Band Generalization for Remote Sensing benchmark reveals that even state-of-the-art foundation models suffer a performance drop of two to four times when generalizing to satellites with no band overlap, and a 5-25% decline when presented with additional bands during inference. This significant degradation underscores the critical need for adaptation mechanisms that can bridge the domain gap between sensors without requiring exhaustive retraining, particularly given the scarcity of labeled data for newer missions.

To address the rigidity of fixed-band architectures, recent research has focused on developing sensor-agnostic models capable of processing arbitrary input configurations. The Any-Optical-Model: A Universal Foundation Model for Optical Remote Sensing introduces a spectrum-independent tokenizer that assigns dedicated band embeddings to each channel, allowing the model to accommodate missing bands or newly introduced sensors while preserving spectral identity. Similarly, Panopticon: Advancing Any-Sensor Foundation Models for Earth Observation extends the DINOv2 framework by treating images of the same geolocation across different sensors as natural augmentations and incorporating cross-attention over channels, enabling effective processing of any combination of optical and synthetic aperture radar inputs. These approaches contrast with traditional methods that rely on fixed input dimensions, offering a pathway toward truly universal remote sensing foundation models. Complementing these architectural innovations, SenPa-MAE: Sensor Parameter Aware Masked Autoencoder for Multi-Satellite Self-Supervised Pretraining incorporates sensor parameters directly into the image embeddings via a versatile encoding module, allowing the model to differentiate between sensors and understand the correlation between sensor specifications and the observed signal.

Beyond architectural flexibility, parameter-efficient adaptation strategies have emerged as vital tools for transferring knowledge across domains with minimal computational overhead. DEFLECT (Deflecting Embeddings for Finetuning Latent representations for Earth and Climate Tasks) proposes an embedding deflection strategy that adapts geospatial foundation models pretrained on RGB imagery to multispectral data by modifying latent representations rather than updating all model weights. This method achieves comparable or superior accuracy to full fine-tuning while utilizing significantly fewer parameters, demonstrating that strong inductive biases from pretrained spatial structures can be leveraged for spectral adaptation. Further enhancing efficiency, recent work on hyperspectral image classification demonstrates that extensions of Low-Rank Adaptation, such as KronA+, can fine-tune multispectral foundation models for hyperspectral tasks with negligible storage overhead, effectively bridging the gap between multispectral pretraining and hyperspectral downstream applications as shown in Parameter-Efficient Fine-Tuning of Multispectral Foundation Models for Hyperspectral Image Classification.

Frequency-based prompt learning and domain generalization techniques offer another avenue for robustness against domain shifts. FrogDogNet: Fourier frequency Retained visual prompt Output Guidance for Domain Generalization of CLIP in Remote Sensing integrates Fourier frequency filtering with visual prompt learning to retain invariant low-frequency components while eliminating noise and background artifacts that vary across domains. This approach has shown superior adaptability in remote sensing scene classification tasks, highlighting the importance of frequency-invariant features for generalization. Additionally, the use of multiple input modalities beyond optical imagery has been shown to improve out-of-distribution generalization. Studies indicate that fusing geographic data layers, such as elevation or land cover maps, with optical imagery significantly enhances model performance in data-limited and geographically distinct settings, suggesting that multi-modal inputs provide a regularizing effect that aids domain transfer as discussed in Using Multiple Input Modalities Can Improve Data-Efficiency and O.O.D. Generalization for ML with Satellite Imagery.

Transfer learning also plays a pivotal role in adapting models to specific sensor constraints, particularly for resource-constrained platforms like CubeSats. For instance, transfer learning from Landsat-7 data to thermal-only CubeSat imagery has enabled accurate cloud segmentation despite the lack of multispectral bands, demonstrating the feasibility of cross-sensor knowledge transfer for specialized onboard tasks as detailed in Transfer Learning for Onboard Cloud Segmentation in Thermal Earth Observation: From Landsat to a CubeSat Constellation. Furthermore,

while models like TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis generate ready-to-use embeddings by fusing optical and radar time series, they still require careful handling of temporal signals to avoid the loss of phenological information common in compositing methods. The collective progress in embedding deflection, frequency-based prompting, and sensor-agnostic architectures indicates a shifting paradigm from training isolated, sensor-specific models toward developing flexible, generalizable systems capable of operating across the expanding ecosystem of Earth observation satellites.

Few-Shot, Zero-Shot, and Training-Free Capabilities

The proliferation of Foundation Models (FM) in remote sensing has fundamentally shifted the paradigm from data-hungry supervised learning to efficient adaptation mechanisms capable of operating in data-scarce regimes. A primary driver of this shift is the ability of these models to leverage self-supervised pretraining on vast unlabeled datasets, thereby acquiring robust representations that require minimal labeled data for downstream fine-tuning. Research consistently demonstrates that in-domain self-supervised learning significantly outperforms generic pretraining when labeled examples are limited, particularly for rare observables where expert annotation is a bottleneck [159]. For instance, models pretrained using Momentum Contrast on domain-specific data surpass ImageNet-pretrained counterparts in low-label scenarios for vehicle detection, addressing the critical scarcity of strategic surveillance data [159]. Similarly, the Prithvi model, pretrained on over 1TB of multispectral data, exhibits accelerated convergence and superior data efficiency across diverse Earth observation tasks, maintaining high accuracy even as the quantity of labeled training data is drastically reduced [62]. This trend is further corroborated by findings in Synthetic Aperture Radar (SAR) analysis, where masked autoencoding pretraining reduces labeling requirements by more than an order of magnitude for vegetation and land cover classification, while also enhancing geographic generalization to regions outside the pretraining distribution [105].

However, the extent of the advantage provided by domain-specific pretraining remains a subject of nuanced debate within the literature. While many studies advocate for the necessity of specialized pretraining, systematic evaluations suggest that the benefits may sometimes be modest or context-dependent. A comprehensive study comparing self-supervised pretraining on Sentinel-2 imagery against ImageNet pretraining found that while domain-aligned pretraining offers competitive performance, the improvements in few-shot settings